



A study of the cooling effect of urban trees: Influencing factors, assessment methods, planning strategies, and impacts

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Abstract

Global climate change and urbanization are causing temperatures to rise in cities, and the resulting heat island effect is becoming increasingly challenging. Urban trees play an essential part in a city's ecosystems by offering a cooling impact through shading and transpiration; this results in reduced urban temperatures, increased thermal comfort for walkers, and an improved urban microclimate. This study integrates the cooling mechanisms, assessment methods, and socio-environmental effects of urban trees, proposing a comprehensive framework encompassing biophysical mechanisms, assessment models, and planning strategies to bridge the gap between research and practice. This review follows the PRISMA systematic review framework and synthesizes 128 eligible studies on the cooling effects of urban trees. Results show that trees reduce air temperature by approximately 0.5–5.8 °C and surface temperature by 2–12 °C, depending on canopy density, climate type, and measurement method. Morphological traits such as leaf area index (LAI > 3) and canopy density were the strongest predictors of cooling in humid and temperate climates, while physiological processes including transpiration enhanced cooling effectiveness under arid conditions. These findings provide quantitative thresholds and a conceptual framework that may support climate-adaptive urban planning and design.

1 Introduction

In recent years, as society has grown, more people have moved to cities. The world urbanization rate has risen to over 56% since the beginning of the twenty-first century, and it is expected to reach 68% in 2050 (Urbanization in the context of global environmental change 2023). The rapid proliferation of urban areas has markedly altered surface coverage and energy equilibrium. Artificial surfaces, including concrete structures and asphalt pavements, exhibit high heat absorption, resulting in elevated temperatures in urban

locales, as compared to their adjacent regions. This intensifies the urban heat island effect (UHI) which, in turn, aggravates energy consumption, degrades environmental quality, and heightens the risk of heatstroke, cardiovascular diseases, and other heat-related ailments among vulnerable populations (Thambawita et al. 2023).

Conventional cooling methods such as air conditioning depend on fossil fuel energy sources, and their use can exacerbate carbon emissions and sustain negative oscillations in urban temperatures. Trees are acknowledged as an economical, sustainable cooling option because of their transpiration, shade, and radiation regulating properties. Their various processes, synergistic with the natural environment, can reduce lower temperatures and can enhance microclimatic conditions. Previous studies suggest that surface temperatures may decrease when tree canopy cover approaches 40%, although the magnitude varies across climate and urban morphology (Coutts et al. 2016a). Research indicates that over 90% of cities across the globe have exhibited an increase in tree cooling effectiveness since 2000 (Zhao et al. 2023), suggesting that systematic tree planning can help enhance cooling efficacy, particularly where water, species selection, and spatial layout are carefully managed. As illustrated in Fig. 1, the conceptual framework of this study

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outlines the relationships among influencing factors, assessment methods, cooling strategies, and socio-environmental impacts.

Tree cooling, as a subject of study within the interconnected domains of planning and environmental science, offers important research value as it encompasses diverse research avenues. A study revealed that, in 83% of cities, trees can lower the monthly maximum temperatures to below 26 °C and can decrease pedestrian-level temperatures by up to 12 °C through sunlight attenuation and transpiration (Li et al. 2024a). In the context of climate change, where extreme heat events are prevalent, urban trees exhibit enhanced cooling efficacy during periods of elevated temperatures, providing additional cooling by shading and transpiration, and thus reducing the risk of high heat during these occurrences (Yang et al. 2024). There is a significant trade-off between tree morphological characteristics and physiological needs. Although a high leaf area index improves the shading effect, it also increases transpiration water consumption. A balance needs to be struck between cooling efficiency and sustainability in planning.

Record-breaking high temperatures in various regions across the globe are resulting in more common extreme weather occurrences; this underscores the critical need for urban cooling initiatives. To mitigate the effects of elevated temperatures, governmental entities at all levels have integrated arboreal cooling into sustainable urban development initiatives. Many areas have launched specialized initiatives to mitigate high heat via tree planting. In 2023, several cities in China launched the Urban Cooling Action Plan aimed at increasing green space coverage through the development of the “City of a Thousand Gardens” and the establishment of a greenway network to alleviate heat island effects, with the goal of achieving a systematic arrangement of tree cooling by 2025. The U.S. Environmental Protection Agency (EPA), on its official website, offers comprehensive guidelines and implementation strategies on utilizing vegetation to mitigate temperatures, assisting regions in exploring environmental adaptation methods to enhance their tree and vegetation cover in order to diminish the heat island effect (U.S. Environmental Protection Agency (n.d.)). The Australian government is advocating for an “urban forestry strategy” to enhance greenery via official policies and programs aimed at lowering metropolitan temperatures and improving environmental quality (Department of Climate Change, Energy, the Environment and Water, Australia ((Department of Climate Change, Energy, the Environment and Water, Australia n.d.))). These policies aim to enhance resilience by implementing systematic greening, conducting scientific assessments, and fostering community engagement, all with

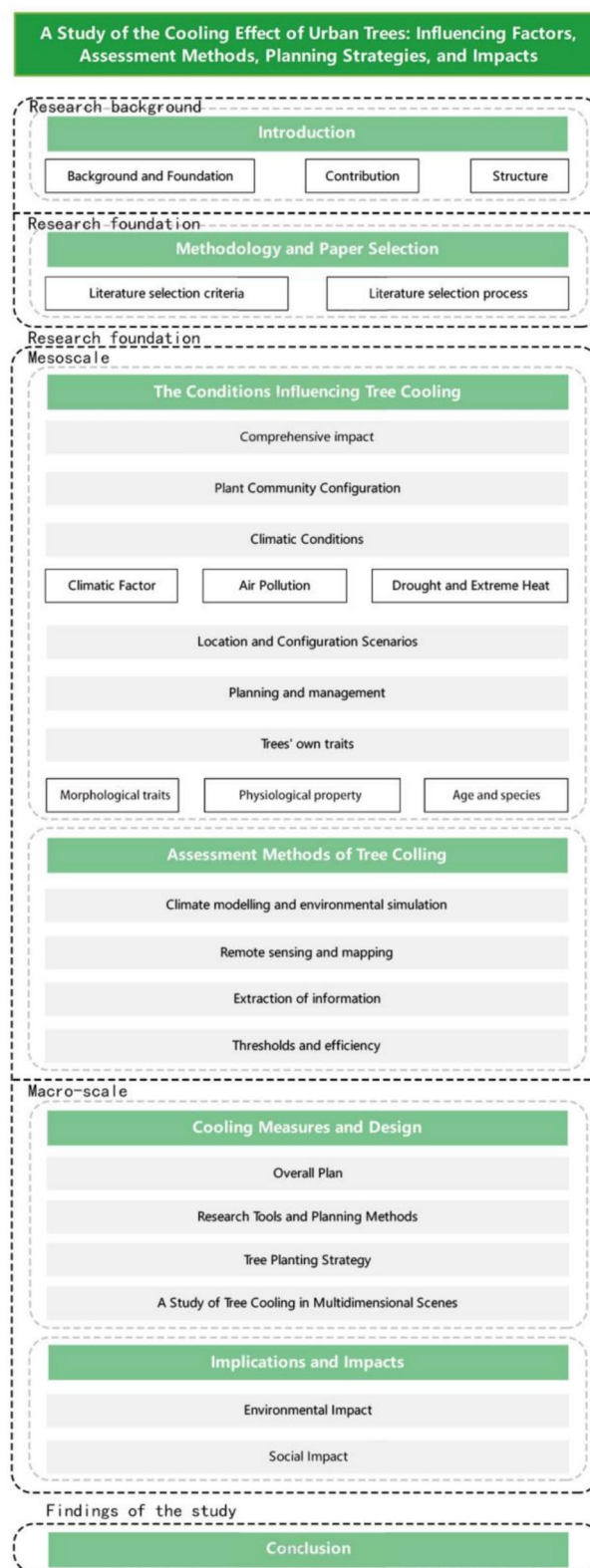


Fig. 1 Article structure

the aim of alleviating heat stress, advancing carbon neutrality, and decreasing mortality rates associated with high temperatures, thereby addressing both present and future environmental challenges.

This study examines the conditions, the research and evaluation methodologies, the specific strategies, and the design frameworks for tree cooling, as well as the effects of tree cooling through a comprehensive analysis of scholarly literature on the subject. By synthesizing the fundamental modules of climatic conditions, arboreal attributes, planning methodologies, and socio-ecological advantages, we systematically examine the determinants of tree cooling, enhance the comprehension of the tree cooling mechanism, and broaden the research domain of urban green infrastructure. The methodologies for researching and assessing tree cooling are categorized into four dimensions: climate modeling and environmental simulation, remote sensing mapping, information extraction, and threshold and efficiency. The methodologies for researching and assessing tree cooling are categorized into four dimensions: climate modeling and environmental simulation, remote sensing mapping, information extraction, and threshold and efficiency. The methods employed in these studies, related to tree cooling, are organized and summarized to serve as a reference for developing the tree cooling research framework. The investigation into tree cooling strategies and design is organized and synthesized to develop a comprehensive methodology that encompasses four components: master planning, research instruments and planning techniques, tree planting strategies, and tree cooling studies across multi-dimensional contexts. This study aims to provide scientific references for urban planning, landscape design, and ecological restoration. The effects of tree cooling are delineated independently from their environmental and social impacts in order to facilitate the formulation of greening policies and sustainable development strategies, including the optimization of street shading and green space arrangement to enhance the cooling efficacy of green areas.

The paper is structured as follows. Section 2 presents an overview of the research methodology and the procedure for selecting the papers. Section 3 offers an in-depth examination of the papers related to factors influencing tree cooling. Section 4 provides a summary of the research and assessment methods related to tree cooling. Section 5 discusses tree cooling measures and design steps. Section 6 evaluates the environmental and social impacts of tree cooling. Section 7 concludes the paper and offers an outlook for future research.

2 Methodology and paper selection

This research into tree cooling encompasses various disciplines and methodologies, considering a substantial body of literature that includes both foundational theories and contemporary studies. This extensive corpus offers a comprehensive overview of the diverse perspectives within the field, aids in identifying research hotspots, and highlights the deficiencies in existing studies. This approach ensures the review's systematic and representative nature, while it also delineates future research directions and provides valuable insights. This systematic review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, implementing a transparent and reproducible procedure encompassing identification, screening, eligibility assessment, and inclusion of studies. The PRISMA flow diagram was employed to maintain consistency and replicability throughout the study selection and synthesis process. All 128 included studies were systematically coded according to study region, methodology, tree-related parameters, and quantitative outcomes to facilitate comparability and enhance analytical rigor (Karimi et al. 2020). The author identified a substantial corpus of research literature on tree cooling across multiple disciplines, but it is marked by inadequate research procedures, insufficient data, and information that is not particularly relevant to the topic of tree cooling. Consequently, It was necessary to establish clear selection criteria to improve the relevance and scientific rigor of the review, to enhance the overall credibility of the review piece.

To ensure methodological robustness and comparability, this review draws upon established frameworks for systematic literature reviews. Recent studies have outlined practical procedures for environmental and urban research synthesis, emphasizing transparency, inclusion criteria, and reproducibility. In particular, the methodological approaches discussed by Linnenluecke, Marrone (Linnenluecke et al. 2020), (Kousis and Pisello 2023), and (Mirabi and Davies 2024, 2022) informed the design of our literature search, screening, and coding process. These references collectively provide guidance on integrating bibliometric techniques with systematic review methodology in the context of climate-related and urban environmental analyses.

Upon reviewing a substantial body of pertinent literature and systematically aggregating and contrasting its research content, methodologies, data sources, and conclusions, it was determined that credible papers in the review of tree cooling systems must exhibit high relevance to the

subject. The literature also needs to exhibit scientific rigor, to demonstrate originality and insight, to reflect timeliness to reveal research trends, and to assert authority with considerable scholarship. Thus, the literature selection process followed these criteria and used a clear and effective technique. While impact factor and citation counts were considered as exclusion criteria for lower-quality publications, this approach follows common practice in environmental sustainability reviews (Karimi et al. 2023). To reduce potential bias associated with journal impact factor, highly cited or regionally representative studies were retained. Initially, it became essential to delineate the research scope concerning the urban tree cooling effect system, in order to establish an interdisciplinary framework encompassing architecture, landscape architecture, planning and design, environmental science, ecology, meteorology, remote sensing, and geographic information science, all while focusing on “tree cooling” and “urban greening.” We additionally examined the keywords “tree cooling”, “urban greening”, “shade impact”, and “transpiration cooling.” The search encompassed the Web of Science, Nature, Google Scholar, and additional databases, and it was conducted comprehensively, in order to guarantee a wide range of data sources. This resulted in the retrieval of 404 pertinent publications. After establishing evaluation criteria, we quickly reviewed abstracts and titles and eliminated 100 publications that were determined to be too broad in scope and were unrelated to tree cooling investigations. We also assessed the impact factor and citation count of the journals in which the articles were published. Moreover, the multidisciplinary analytical framework adopted in this review aligns with recent methodological discussions on integrated urban climate and environmental research (Ecological footprints of climate change 2022), reinforcing the cross-domain synthesis of physical, ecological, and social dimensions. Ultimately, we excluded 176 low-quality articles. Consequently, we compiled a set of 128 eligible literature pieces, to ensure the innovation, rigor, and representativeness of the selected works. The literature that fulfills the criteria was categorized based on research perspectives such as influencing conditions, methodologies, cooling measures, and impacts. A systematic extraction of the study methodologies, results, and innovations from each piece of literature was performed, accompanied by continuous omission checks and dynamic updates, in order to enable thorough analyses and discussions. The overall process of literature identification, screening, and inclusion followed the PRISMA framework, as shown in Fig. 2.

In addition to simulation and remote sensing approaches, recent studies have applied statistical modelling techniques to quantify and interpret spatial variations in tree cooling. Generalized Additive Models (GAMs), for example, have been proven effective for testing nonlinear relationships and

identifying region-specific drivers of cooling performance. Integrating such statistical models with simulation outputs or remote sensing data allows for more robust estimation of tree cooling effects and their environmental determinants. This methodological combination broadens the analytical scope of urban microclimate studies and enhances the capacity to detect significant spatial heterogeneity in tree-induced cooling (Mirabi et al. 2025).

3 The conditions influencing tree cooling

The research on tree cooling predominantly focuses on the variables that affect this phenomenon, encompassing environmental configurations, the inherent characteristics of the trees, and a review of influencing elements and other research avenues. The cooling capacity of urban trees is influenced by multiple interacting factors, including tree traits, plant community structure, climatic conditions, spatial arrangement, and green space management. These factors do not operate independently, and their combined effects determine actual cooling outcomes (Yang et al. 2024). As illustrated in Table 1, the cooling capacity of urban trees is influenced by a complex interplay of factors, which this review synthesizes into several key categories: tree traits (morphological and physiological), plant community configuration, climatic conditions, location and configuration scenarios, and planning/management contexts.

Morphological traits—including height, crown width, leaf area index (LAI), canopy density, root depth, and branching pattern—primarily influence shading and solar radiation interception. Trees with larger crowns and higher LAI can lower local air and surface temperatures by 1–3 °C through increased shading and reduced shortwave radiation exposure, while deeper roots enhance growth stability and long-term shading potential (Sharmin et al. 2023).

Physiological traits, such as stomatal conductance, transpiration rate, photosynthetic rate, respiration, and water-use efficiency, regulate evapotranspiration and latent heat flux exchange. Species with higher transpiration and photosynthetic rates demonstrate greater evaporative cooling capacity, whereas those with high water-use efficiency can sustain cooling effects under drought or heat stress. Field observations indicate that trees with high stomatal conductance can reduce ambient air temperature by approximately 0.5–2 °C compared to species with lower transpiration capacity under comparable conditions (Wei et al. 2021). In terms of community structure, variables such as canopy density, hierarchical structure, and species diversity can considerably affect microclimate regulation. Multi-layered and diversified vegetation communities enhance the overall cooling benefit by strengthening shade–transpiration

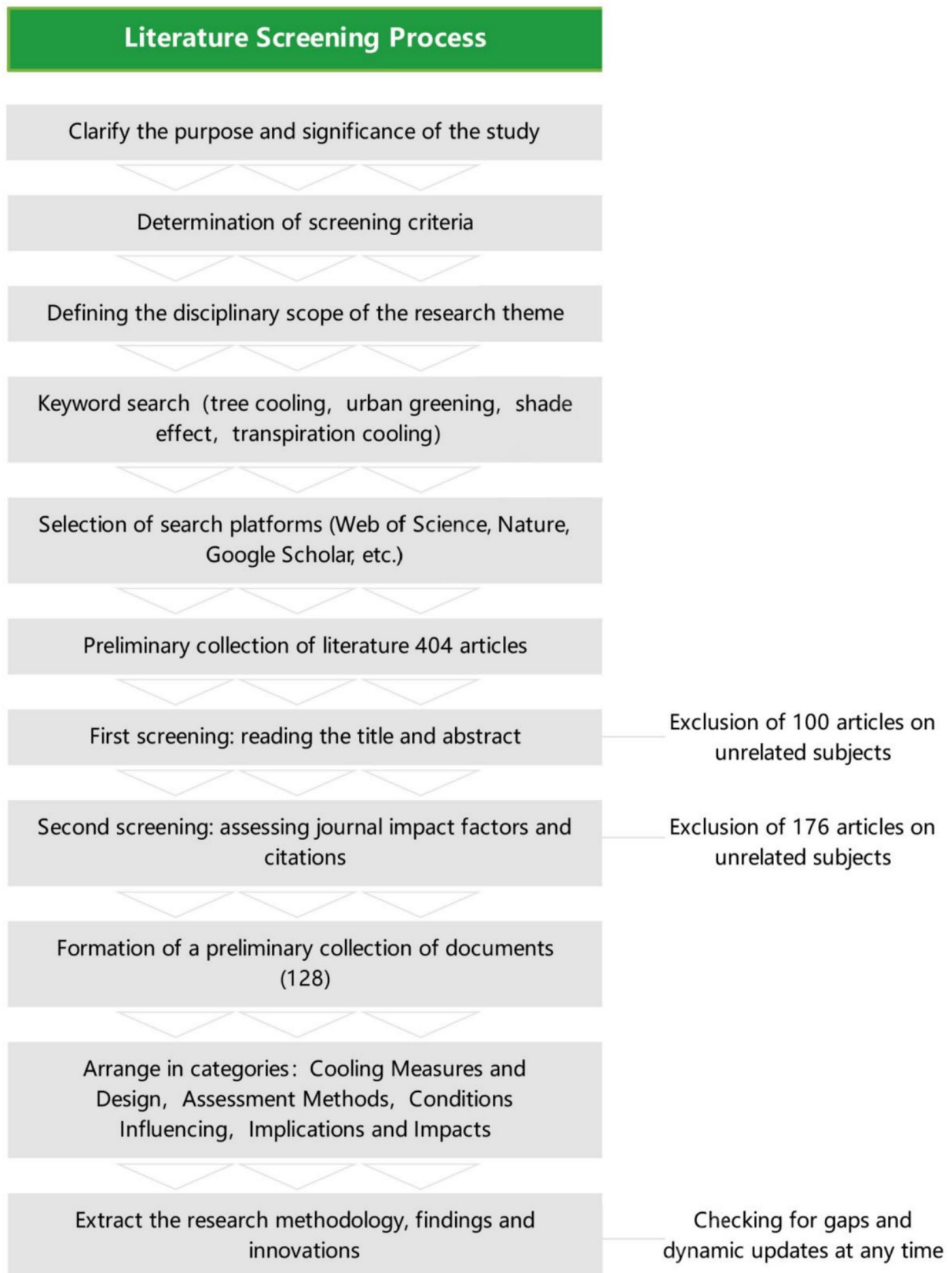


Fig. 2 Literature Screening Flowchart

Table 1 Classification of tree-related factors affecting cooling performance

Category	Key Variables	Cooling Mechanism	Quantitative Indicator (Range)
Morphological	Tree height, canopy width, LAI	Shading	2–6 °C air temperature drop
Physiological	Transpiration rate, stomatal conductance	Evapotranspiration	1–3 °C humidity-driven cooling
Species	Broadleaf, coniferous, deciduous	Combined	0.8–2 °C difference
Age	Young, mature	Shade & water efficiency	20–30% variation in CE

synergy (Mahmuda, et al. 2023). Regarding climatic conditions, factors including ambient temperature, air humidity, wind speed, solar radiation intensity, pollution levels, and weather extremes can alter tree cooling efficiency (Yan et al. 2023). The quantitative relationships between key climatic variables and tree cooling efficiency are summarized in Table 1.

At the species level, studies have revealed distinct variations in cooling efficiency associated with morphological and physiological traits. Broadleaf deciduous species generally exhibit stronger cooling effects than coniferous or evergreen species due to their higher leaf area index, greater transpiration rates, and lighter canopy albedo. In urban streets and parks, empirical studies also demonstrate that tree species with larger canopies and higher stomatal conductance achieve greater microclimate regulation efficiency. These species-level comparisons highlight the importance of selecting tree species based on both canopy structure and physiological adaptability to local climatic conditions (Wei et al. 2021).

Spatial arrangement, comprising the positioning of tree plantings and surface materials with planting indices such as sky view factor and enclosure area, influences tree cooling by enhancing shade surfaces, cumulatively reducing the sky view factor, and elevating surface albedo (Yan et al. 2023). And green space coverage and the rational selection and layout of tree species urban green space planning and management elements also have a major impact on tree cooling (Yang et al. 2024). The key environmental and structural factors influencing tree cooling are summarized in Fig. 3, which illustrates how climate, tree traits, and spatial configuration interact to determine cooling outcomes.

3.1 Comprehensive impact

This review examines the conditions that impact tree cooling, as it performs a thorough analysis of the environmental factors influencing urban trees and the effect of tree growth characteristics on microclimate regulation. It evaluates the relationship between these variables and the cooling effect by measuring tree growth-related parameters and environmental factors both within and outside urban forests, while it systematically analyzes the impact of tree and environmental factors on cooling to inform urban forest management planning (Wang et al. 2018a). The attributes of street trees and microclimatic variables influence the air temperature ΔT , which exhibits a positive correlation with leaf area index, canopy width, and solar irradiance, while demonstrating a negative correlation with wind speed and water vapor pressure difference. The findings of this study also offer theoretical support for the strategic incorporation of trees into landscape planning for cooling purposes (Sharmin et al. 2023). This is also a comprehensive analysis of 182 studies on tree cooling under 17 climatic conditions in 110 cities or regions around the world, which concluded that urban form and climatic conditions jointly determine the effectiveness of tree cooling and, to cope with climate issues, examines the regulation of maximum and pedestrian-scale temperatures in different climatic zones and humidity environments in order to provide greening guidance on the use of tree cooling (Li et al. 2024). It also includes analyzing the coupling and decoupling of the urbanization gradient with different forest types by measuring differences in climatic conditions, tree size, and composition of the sample plots, as well as clarifying the effects of climate, tree size, and species composition on microclimatic regulation using Redundancy Analysis (RDA) and Relative Importance Assessment (RIA) to maximize the cooling effect of trees (Wang et al. 2019).

To support heat mitigation planning for trees in urban landscapes, the influence and the association of tree characteristics and microclimatic environmental conditions on sub-canopy air temperatures (ΔT) of street trees can be evaluated separately by tracking the sub-canopy air temperatures of various tree species as well as the characteristics of trees and tree canopies (Wei et al. 2021). The planting index can also be used to provide a more thorough and focused investigation of the variables influencing tree cooling. Three planting indices—sky view factor, leaf area index, and enclosure area—were found to have an impact on the thermal environment when correlated with indicators of the thermal environment. For example, a higher sky view factor would result in a higher temperature, while a lower leaf area index and a larger enclosure area would also have

Condition of Influence	Comprehensive Impact	Environmental Factors and Tree Growth Characteristics of Urban Trees			
		Street tree tree characteristics and microclimate variables			
		Tree characteristics and microclimatic conditions			
		Tree traits, urban form and climatic conditions			
		Urbanisation gradient and forest types			
		Plantation Index			
	Plant Community Configuration	Different Forest Types and Structures			
		Different Vegetation Measures			
		Characterisation of Tree Community Canopy Structure			
		Tree Canopy Ecological Niches			
		Community Structure			
		Biodiversity			
	Climatic Conditions	Climatic Factor	Weather Differences		
			Seasons and Weather		
			Climatic and Environmental Factors		
		AirPollution	Ozone Pollution		
		Drought and Extreme Heat	High Temperatures and Drought		
			Droughts		
			Scorching Heat Wave		
	Location and Configuration Scenarios	Skyscape Factor			
		Surface Albedo			
		Different Tree Configurations			
		Planting Location			
	Planning and Management	Street Canyon Planning Factor			
		Park Planning Factors			
		Urban Form			
	Trees' Own Traits	Morphological Traits	Crown Structure and Leaf Morphology	Crown Cover, Size	
				Vertical, inner and outer canopy structure	
				Shade and transpiration processes in the canopy	
				Crown Properties	
				Leaf Area	
			Vertical tructure of trees	Crown volume and diameter at breast height	
				Tree species, tree height, crown base height and trunk circumference	
				Leaf area index, crown spread and tree height	
			Shade	Durability	
				Humidity Type	
				Vertical Dimension	
				Trends	
				Time Dimension	
			Physiological Property	Transpiration	
				Leaf Area Density	
		Stomatal Conductance			
		Age and Species	Tree Type		
Tree Species					
Age of Trees					

Fig. 3 Factors affecting tree cooling

a thermal effect. This suggests that the heat island effect can be effectively reduced by optimizing the greening design and, thereby, improving the thermal environment (Guo et al. 2022).

Collectively, these studies demonstrate that tree cooling is not dictated by a single factor but emerges from the interaction of tree traits (e.g., LAI, canopy width) and ambient conditions (e.g., solar radiation, wind speed). The most effective cooling occurs when high-leaf-area trees are positioned in areas of high solar exposure and low wind, optimizing the synergy between shading and transpiration.

3.2 Plant community configuration

Research on the impact of plant community configuration on tree cooling includes a related study that evaluated the impact of tree species diversity on street tree cooling using biomass reciprocity, shading mechanisms, and structural diversity analyses. The study combined forest data with local site satellite meteorological data, average air temperature, and canopy cover. It concluded that the more diverse the tree species, the better the trees' daytime cooling effect would be (Rendon et al. 2024). Through the measurement of plant community characteristics like canopy area and density and by measuring microclimate characteristics like air temperature, solar radiation, wind speed, and relative humidity at six locations in a semi-arid city, related articles also found that multi-layered communities enjoy the best cooling effect and that canopy area, density, tree height, and solar radiation all affect cooling efficiency. The researchers also proposed a calculation model to assess the impact of community structure on climate comfort; it offers theoretical and technical support for urban management (Yang Yang et al. 2019).

The research examined the impact of plant community canopy on tree cooling, utilizing UAV LiDAR to assess the diversity and complementarity of tree canopy ecotypes. The analysis focused on their effects on the vertical microclimate of urban forests. Findings indicated that the vertical temperature variation in urban forests is affected by the diversity of canopy ecotypes (strucPCA1 and strucPCA2), and complementarity (GSI and GSCI), as well as the leaf area index. Conclusions were drawn to formulate strategies for optimizing the vertical microclimate of urban forests through the enhancement of canopy structure and ecological management practices (Jia et al. 2024). Meteorological data from various plant communities in urban areas were collected through field experiments to analyze the relationship between three-dimensional structural features and cooling benefits. The findings indicated that the tree layer influences cooling, and this provides a scientific basis for optimizing green space planning (Yu et al. 2024).

In the study on the effect of the combination of vegetation types on the cooling of trees, through the computational fluid dynamics (CFD) simulation of the airflow of the locally applied vegetation in the street, the RANS model was used to verify that the street trees have the largest contribution to the cooling of the pedestrian layer (with a cooling rate of 2.5–3.2 °C), and the effectiveness of the greening of the façade and the rooftop was found to be less than 30%. It was found that the synergistic effect of multiple greening measures has spatial limitations, that the cooling amplitude is regulated by both vegetation type and spatial layout, and that it is necessary to design a composite greening scheme to optimize the efficiency of heat island mitigation (Gromke et al. 2015).

At the same time, some articles have also assessed the microclimate effect of tree communities by studying the canopy characteristic index and discomfort index, and they have found that tree communities reduce temperature, humidity, and light intensity and improve human comfort compared with open space (i.e., areas devoid of tree cover, such as paved plazas or open lawns, which are fully exposed to solar radiation). Canopy structure characteristics (depression, i.e., the degree of openness or gaps within the canopy that allows for light penetration and air movement, leaf area index) are the key regulating factors; these provide a scientific basis for the optimization of urban greening layout and screening of cooling tree species, and they support the ecological strategy of heat island mitigation and comfort synergy (Yan et al. 2012).

As a collection of tree-plant community configuration, the study of the impact of different forest types and structures on climate also provides value for the study of the impact of plant community configuration on tree cooling. By comparing the climatic conditions of under-canopy and open sites, the study found differences between under-canopy and open sites in indicators of temperature, humidity, wind speed, and photosynthetically active radiation (PAR), and that such differences were closely related to forest type and topographic features. Forest type dominated the PAR and maximum temperature variations, while minimum temperature was influenced by both forest type and slope orientation. Forest structure and topographic factors play a key role in regulating local microclimate, and their seasonal differences have an important influence on temperature and humidity changes (Renaud et al. 2011).

A comparative insight across these studies is that complex, multi-layered plant communities consistently provide superior cooling compared to monocultures. This underscores a key principle: biodiversity in structure and species not only enhances ecological resilience but also amplifies the microclimate regulation service of urban greenspaces.

3.3 Climatic conditions

3.3.1 Climatic factor

Investigations on the impact of climatic conditions on arboreal cooling encompass analyses of climatic variables, air pollution, drought, and extreme temperatures. Investigations on the influence of climatic variables on arboreal cooling encompass analyses of meteorological variations, seasonal changes, and ambient climatic factors. Investigations into the impact of meteorological variations on tree cooling indicate that atmospheric conditions primarily influence the efficacy of tree cooling and that a notable synergy exists between the configuration of street canyons and the morphology of trees. The findings indicate that the impact of meteorological variations on microclimate control far surpasses the effects of static urban morphology and forest attributes among various cooling mechanisms such as air and soil cooling, shading, and humidification. A dynamic control technique that optimizes tree architecture using real-time meteorological data and that emphasizes enhancing the combined effects of canopy shading and transpiration in heat island regions is proposed (Wang et al. 2018b).

A study examining the impact of seasonal and meteorological factors on tree cooling continuously monitored conditions in four peri-urban woodlands and one roof control point within a humid subtropical climate. The findings revealed that the cooling effect of trees in these woodlands exhibited notable seasonal and diurnal variations, with solar radiation and ambient air temperature serving as the primary

regulating factors. Dynamic monitoring utilizing air temperature, Physiological Equivalent Temperature (PET), and Universal Thermal Climate Index (UTCI) indicated that daytime radiative heat load and nocturnal air temperature variations collaboratively influenced alterations in cooling amplitude. The study advocated for the implementation of a continuous monitoring system throughout the seasons to measure the temporal coupling mechanism between climatic variables and canopy shading (Cheung et al. 2020).

The research examining the impact of climatic conditions on tree cooling employed an ordinary least squares (OLS linear) regression model to analyze variations in tree cooling efficiency (CE) attributable to alterations in vegetation density. Notable geographic disparities in urban CE were identified, characterized by climatic factors that delineate a hydrothermal gradient, with temperature prevailing in humid regions and humidity sensitivity evident in arid areas. A cross-biome regression study validated the synergistic regulatory influence of wind speed, indicating that greening must be tailored to improve transpiration in arid regions and to optimize shade in humid areas to boost heat mitigation effectiveness (Wang et al. 2020). Table 2 synthesizes findings on how community structure, canopy density, and species diversity affect local microclimate regulation and cooling intensity.

3.3.2 Air pollution

Research examining the impact of air pollution on tree cooling encompasses a study on the impacts of ozone pollution

Table 2 Effects of climatic factors on tree cooling

Climatic Factors Affecting Tree Cooling	Main Cooling Mechanisms	Standard of Measurement	Research Methodology	Conclude
Air Temperature (T_{air}), Relative Humidity (RH), Light Intensity	Climatic conditions are the main factor (65.4 per cent unique contribution)	Fig. Local Environment	Field Survey, Linear Regression, MANOVA, RDA	Cooling effect varies with urban–rural gradient
Incident Shortwave Radiation (S_{in}), Temperature Air (T_{a}), Relative Humidity (RH), Wind speed (v), Rainfall, Soil Moisture (SM), Vapour Pressure Deficit (VPD), Net Longwave Radiation at Night ($L_{\text{net_night}}$)	Seasonal and meteorological changes	Urban Thermal Climate Index (UTCI), Sky Visibility Factor (SVF)	Control Monitoring, Multiple Linear Regression Analysis	The magnitude of cooling varies with seasonal and meteorological conditions, with an average cooling of 0.9 ± 0.5 °C (T_{a})
Air Temperature, Humidity, Wind Speed	Air temperature and humidity affect Cooling Efficiency (CE)	Urban Tree Proportion (P_{tree}), Land Surface Temperature (LST)	Statistical Analysis, Bivariate Relationships, Partial Correlation, Multiple Regression Analysis	CE is higher in hot dry biomes and broadleaf trees cool better than conifers
Summary	Context-dependent	Multi-metric	Field & statistical	Varies by climate, season & species

on this phenomenon. The study measured the leaf area index (LAI) of trees in two regions by utilizing the ENVI-met model and biometeorological data. It revealed that ozone pollution substantially diminished the cooling effectiveness of trees, resulting in sensible heat accumulation in open areas and intensifying heat stress in regions with low canopy density. Urban greening design must integrate pollution exposure with canopy structure optimization to develop a three-dimensional plant system resilient to ozone stress (Wai et al. 2020).

3.3.3 Drought and extreme heat

A greater number of research studies have investigated the impacts of drought and exceptionally high temperatures on tree cooling. Concerning the diurnal cooling capacity of trees during drought conditions, an analysis of vegetation cover in urbanized regions that used satellite and airborne thermal imagery revealed that the thermal regulation of vegetation under drought stress exhibited considerable variation; tree-covered areas sustained a stable thermal buffer, whereas grasslands encountered a marked decline in cooling efficacy. Consequently, a tree-prioritized green space resilience network should be established, accompanied by an intelligent watering system to improve canopy transpiration-shade synergy during periods of drought. (Allen et al. 2021).

Relevant studies encompass an investigation into the impact of spring heat waves on arboreal cooling. An analysis of urban tree growth data in relation to cooling efficiency (CE) revealed that the cooling effect of trees in regions with substantial vegetation cover during warm spring heat waves primarily relied on water-use efficiency, while variations in temperature, precipitation, and wind speed predominantly influenced the interannual fluctuations of CE. To address the difficulties of climate change, enhancing greening in low-coverage areas is advised, to optimize the cooling efficacy of trees through improved water management (Wei et al. 2023).

The decoupling of stomatal conductance and photosynthesis in trees at exceptionally high temperatures also influences tree cooling. Analysis of recorded leaf gas exchange across various tree species revealed that vegetation transpiration exceeded the model estimates in adequately hydrated regions, with a notably pronounced increase in drier locations. The current surface models lack the integration of this adaptive mechanism, potentially resulting in a systematic underestimation of evapotranspiration predictions in high-temperature and high vapor pressure deficit environments. Consequently, model parameters require optimization to enhance the accuracy of eco-hydrological simulations (Marchin et al. 2023).

The research on tree cooling in reaction to drought and intense heat has examined the impact of tree cooling across various climatic zones, utilizing high-resolution satellite surface temperature (LST) and European multi-city land cover data. The findings indicate that trees can markedly lower surface temperatures in summer and during extreme heat events, whereas the cooling efficacy of treeless green areas is considerably inferior to that of tree-covered environments. Urban design should prioritize enhancing tree planting to bolster high-temperature mitigation capacity (Schwaab et al. 2021).

Measurements of sap flux density and leaf gas exchange for four prevalent tree species in parks, forests, orchards, and urban streets in high-latitude cities, particularly in Northern Europe, indicated that sap fluxes increased during heat waves, while evaporative demand-driven fluxes generally rose during droughts. Trees provide cooling functions during heatwaves and droughts, and preservation of these cooling services through moisture threshold monitoring and the optimization of drought watering is necessary (Ahongshangbam et al. 2023). Although morphological and physiological factors have often been examined separately, their interactions are critical for understanding the actual cooling performance of trees. A trade-off frequently emerges between morphological traits such as high leaf area index (LAI), which enhances shading efficiency, and physiological demands such as high transpiration rates, which increase water consumption. This interplay suggests that planners should balance cooling effectiveness with resource sustainability, particularly when selecting species for arid or water-stressed regions.

Elevated temperatures and drought conditions positively influence the cooling effectiveness of trees. The study assessed the reaction of urban trees to increasing temperatures using data on their cooling efficiency from 2003 to 2019, revealing that warming greatly improves cooling efficacy. The combined regulation of temperature, humidity, and precipitation produced enhancement in cooling efficiency in warm, humid cities. This mechanism is linked to climate factors' capacity to influence cooling via transpiration efficiency, offering a scientific foundation for optimizing tree distribution to alleviate thermal risks associated with global warming (Yang et al., 2024).

The influence of climate is paramount, often overriding other factors. A critical comparative finding is that trees in arid regions rely more on physiologically-driven transpirational cooling, while in humid regions, morphologically-driven shading dominates. This dictates that species selection and water management strategies must be climate-specific to maximize cooling efficiency.

3.4 Location and configuration scenarios

Research on the impact of external environmental factors on tree cooling encompasses the effects of location and configuration situations on tree cooling. The research on the impact of planting location on tree cooling has revealed that central planting on the leeward side (i.e., the downwind side) yields superior cooling effects, as compared to a uniform distribution. This is because a concentrated tree cluster on the leeward side can more effectively create a sizable, sheltered cool zone. The windward side, being more exposed, allows cool air generated by trees to be quickly advected away, while the leeward side benefits from the accumulation and downwind transport of this cooled air, thereby extending the cooling influence. This conclusion was drawn from empirical measurements of heat stress at the park site and from rapid simulation analysis. Consequently, the study proposes a method for identifying planting locations as well as a planning strategy to maximize cooling benefits under resource constraints, while it also offers landscape design and research aimed at achieving the park's cooling objectives optimization plan (Hao et al. 2023).

The impact of street tree arrangement and positioning on the roadside thermal environment in a tropical urban canyon has been analyzed using the ENVI-met 3D microclimate model for simulation purposes. The results demonstrate that, in hot and humid areas, dense, continuous rows of trees with high leaf area density (LAD) reduce ambient temperatures, whereas widely spaced rows of high-LAD trees are more effective in enhancing thermal comfort in hot and humid climates. The study indicates that strategic design in tree planting, including adjustments in density and leaf area density, is more effective in optimizing the microclimate compared to ad hoc methods, thus providing a scientific basis for street vegetation planning (Jayasinghe et al. 2024).

The arrangement of individual trees in small green space designs can affect their cooling impact. The study concluded that mobile air temperature monitoring data, along with the classification of small green space types and structures, indicate that the cooling efficacy of trees in a polygonal mixed vegetation arrangement is optimal and that the planting position affects the cooling effect. The conclusions provide a basis for improving the planning of neighborhood green spaces, advocating for the prioritization of mixed vegetation arrangements and the strategic placement of tree locations to enhance the cooling service capacity of street greens (Park et al. 2017).

The sky view factor is an external environmental criterion that influences tree cooling. The analysis of gridded measured climate data from urban parks has revealed that an increase in the park sky view factor (PSVF) correlates with both enhanced cooling intensity and a reduction in

mean radiant temperature (MRT). While high-density trees can disrupt airflow, an increase in PSVF enhances the overall thermal environment. Optimizing tree and canopy layout is advised, in order to improve outdoor thermal comfort in subtropical coastal cities. (Zhang et al. 2019).

The external factor of surface albedo influences tree cooling. An article investigated the effects of tree planting and increased surface albedo (solar reflectance) on near-surface object temperatures and air temperatures, focusing on the UHI effect during summer. The study found that near-surface temperatures can be effectively reduced only when the albedo surpasses a critical threshold. Additionally, the net cooling effect of trees through shade and transpiration is moderated by the extent to which the canopy dissipates heat. The overall cooling impact of shading and transpiration is influenced by the extent of mixing between warm air trapped within the urban canopy layer and the cooler air near the ground surface. The cooling effect on near-surface air temperatures is minimized when approximately 45% of the canopy-heated air is mixed with the ground, indicating variations in the conditions applicable to different strategies (Levinson 1997).

The impact of variations in surface paving and surface albedo on tree cooling has been examined in an article that analyzes the physiological, growth, and cooling benefits of three common tree species with differing drought tolerances. The findings indicate that the growth, physiological functions, and cooling benefits of trees in densely paved areas are constrained, with this limitation surpassing the differences in drought tolerance among the tree species. High drought-tolerant tree species are recommended for planting in areas with low paving density, whereas high-density paved areas should serve non-core shade functions. This approach provides a scientific basis for optimizing tree planting design in the context of climate change (Konarska et al. 2023).

These findings highlight that even an ideal tree can have a diminished effect if placed incorrectly. The comparative analysis reveals that cooling efficacy is highly sensitive to spatial context; for example, the same tree species provides greater thermal comfort improvement in a shallow street canyon or a central park location than in a deep, already-shaded canyon.

3.5 Planning and management

Both the planning and the management of the surrounding environment influence tree cooling, encompassing factors such as park design, street canyon configurations (i.e., the geometric arrangement of buildings lining a street, which defines its aspect ratio, orientation, and sky view factor, thereby critically influencing solar access, wind flow, and

heat trapping), urban morphology (i.e., the overall physical structure and pattern of a city, including the spatial arrangement of buildings, streets, green spaces, and other land uses, which shapes mesoscale climate processes), and additional considerations. This section synthesizes findings from diverse studies to identify common planning principles and management strategies that enhance tree cooling performance across different urban contexts. The research on the influence of park planning factors (e.g., the spatial configuration of vegetation, the presence and size of water bodies, the geometry of open lawns, and the overall landscape design) on tree cooling indicates that these factors, particularly tree cover, affect thermal environment indicators. This conclusion is drawn from the measurement and collection of climate data in urban parks, alongside the application of multiple linear regression analysis on the thermal indicators related to park planning factors, offering important insights for urban planning (Zhang et al. 2021).

The research examined the effects of planning factors on tree cooling within street canyons, focusing on the influence of street trees on the thermal environment. This was achieved by measuring both meteorological parameters and the thermal climate index (UTCI) across three east–west street canyons. The findings indicate that in shallow canyons, street trees can markedly lower temperatures and mitigate thermal stress, whereas deep canyons exhibit limited cooling capacity due to the shading effects of buildings. Street geometry, tree cover, and meteorological conditions collectively influence cooling effectiveness. It is recommended that street tree planning optimize planting density and species selection in conjunction with the spatial characteristics of the canyon to improve the thermal environment adaptability of urban streets (Coutts et al. 2016b).

The influence of urban form planning and management on tree cooling was observed to mitigate the heat island effect, particularly through radiant heat during summer and winter nights in urban settings. This was assessed by measuring air temperature, relative humidity, and wind speed at various locations within urban areas in China. Trees lower temperatures during both day and night in summer, attributed to their extensive leaf area and considerable transpiration. During winter, while cooling is restricted, this decreases wind velocity and improves thermal comfort. The design of urban form and environmental planning impacts tree cooling and enhances outdoor thermal comfort in temperate climate zones (Tong et al. 2017).

The planning and management factors, from park design to street canyon geometry, act as a macro-scale framework that can either amplify or constrain the cooling benefits of individual trees. This underscores that effective tree cooling requires coordination across multiple scales of urban governance, from landscape architecture to municipal

zoning. Across the reviewed studies, effective planning and management practices share common strategies—integrating vegetation design with urban form, optimizing species selection to local climates, and combining spatial planning with maintenance frameworks—to achieve scalable and resilient cooling benefits.

3.6 Trees' own traits

3.6.1 Morphological traits

(1) Crown structure and leaf morphology

The inherent characteristics of trees influence their cooling effects, with morphological traits encompassing canopy structure, vertical structure, and shade, among others. Physiological traits encompass leaf area, stomatal size, and other factors. Additionally, variations in species and the age of the trees will influence the efficiency and the outcomes of tree cooling. This study examines canopy structure and leaf morphology, focusing on how trees' morphological characteristics influence cooling. It specifically investigates the impact of canopy structure, including leaf number, density, and angle, on tree cooling effects. The research assessed the horizontal infrared radiation performance of the canopy, revealing that variations in infrared transmittance and reflectance were primarily influenced by overall structural characteristics, including leaf count, density, and orientation, rather than individual leaf traits. Canopy transmittance exhibited variation across four directions and at different times of day, thereby confirming the critical influence of structural characteristics on the regulation of radiant heat. Urban greening should prioritize canopy morphology design to optimize microclimate regulation and to enhance thermal comfort (Deng et al. 2019).

Several studies have examined the relationship between urban tree canopy cover (UTC) and the thermal environment using multi-source remote sensing and statistical analysis. These studies confirm that tree canopy cover lowers urban temperatures, with the expansion of the patch area being more crucial for cooling efficiency than an increase in circumference. The findings highlight the importance of optimizing the spatial scale of tree canopy to reduce the heat island effect and improve urban livability, offering a scientific foundation for the development of sustainable forest cities in comparable areas (Dai et al. 2023). The vegetation canopy structure and its three-dimensional point cloud volume (L_V -3DPC) were derived using terrestrial laser scanning (TLS) technology. This data, in conjunction with meteorological information, facilitated an analysis of the cooling effects of shade and transpiration on the forest

floor. Results indicated that small-leaved species exhibited superior cooling efficacy, when compared to large-leaved species. Furthermore, the canopy structure, leaf area index, and sky view factor were found to influence cooling efficiency. The analysis demonstrates that shade provided by tree canopies is the primary factor in the cooling process, introducing a novel TLS-based method for analyzing canopy shading effects on surrounding air and surface temperatures. This approach offers technical support for selecting small-leaved tree species and optimizing canopy structure design to improve both thermal environment control and energy efficiency (i.e., reducing energy consumption in adjacent buildings primarily by lowering cooling demand through shading and transpirative cooling) in urban green spaces (Kong et al. 2016).

The study examined the thresholds of structural parameters, including tree canopy density, porosity, and vegetation type, for cooling and humidification using a dummy variable regression method. Canopy density exhibited a positive correlation with cooling and humidification, while porosity facilitated cooling but hindered humidification. The influence of various vegetation types on ecological benefits showed variation, and urban green coverage, along with spatial layout, also markedly impacted cooling and humidification effects. Moving forward, these findings offer theoretical insights and practical guidance for the plant configuration and temperature regulation functions of green spaces (Wei et al. 2021). Research evaluated the cooling effect of individual and clumped trees using a canopy density model, incorporating actual measurements of tree canopy density and the computer simulation tool ENVI-met. Findings indicate a positive correlation between canopy density and cooling effect, with high-density tree species (e.g., ficus) demonstrating greater efficacy in reducing ambient temperature. Additionally, clumped planting enhances the cooling effect, and optimizing tree density and planting groups can further improve temperature regulation capabilities (Wei et al. 2021). Optimizing tree density and planting configurations can enhance the cooling effect and achieve microclimate regulation (Shahidan 2015).

The size of a tree's crown influences its cooling effect. A study employing infrared thermography to measure the surface temperatures of various urban surfaces beneath the shade of different tree species, in comparison to unshaded areas, revealed a strong correlation between the cooling effect and the leaf area index as well as canopy coverage. Notably, coniferous species exhibit a pronounced cooling ability; however, their crown size restricts the extent of shade provided. Multiple regression models have demonstrated that single tree planting contributes to mitigating the heat island effect. Additionally, optimizing canopy structural parameters, such as leaf area index and canopy

width, offers theoretical support for tree species selection and canopy design in urban planning (Speak et al. 2020). Several studies have investigated the impact of tree cover and canopy size on the thermal environment by comparing field-measured temperature data with ENVI-met simulation data to assess the reliability of ENVI-met. "The study found that cooling effect increased progressively with greater tree cover, and that under equivalent cover small-canopy patches sometimes produced stronger near-surface cooling than larger continuous canopies. This pattern has been attributed to interactions between shading and transpiration processes: Jiao et al. (2017) demonstrated that patch size and canopy configuration modulate the relative roles of shading and evapotranspiration, and that certain smaller or fragmented canopy arrangements can enhance airflow penetration and latent heat flux per unit leaf area, thereby increasing local cooling efficiency under some conditions."

For the study of the effect of canopy on tree cooling in subtropical regions, through the use of field measurements and questionnaires, it was found that the ratio of crown diameter to branch position height (D/PH) had an effect on tree cooling and that there was a strong correlation between D/PH and the average radiant temperature and thermal sensory polling, whereas canopy porosity had a lesser effect on tree cooling. This provides important reference for urban planning and landscape design, especially in terms of plant selection for improving the urban microclimate (Zhang and Gou 2021).

Canopy structure affects the vertical temperature structure of trees, and the study found that canopy temperature was correlated with air and surface temperatures by measuring the differences in canopy, air, surface and soil temperatures of three forest types, suggesting that canopy temperature can effectively predict air and surface temperatures. Meanwhile, it was found that ambient temperature and humidity, solar radiation, and canopy structure (thickness, species, and tree height) affected the forest thermal environment and cooling (Dong et al. 2022). It has also been found that, by using field measurements of two types of street trees, air temperature could be recorded at different heights while meteorological and soil variables were monitored. It was also found that transpiration was stronger in trees with larger leaf areas, that the temperature gradient from the outside to the inside of the canopy decreased more, and that as air temperature increased, soil moisture decreased, temperature increased, and cold air from the canopy mixed with sensible heat fluxes, resulting in a reduced cooling effect of the trees in extreme hot weather. It was concluded that deep canopy shading plays an important role in tree cooling (Rahman et al. 2018). Table 3 lists the major climate modelling and simulation tools applied in tree cooling research, highlighting their respective spatial scales and strengths.

Table 3 Effects of canopy structure and leaf morphology on tree cooling

Research	Research Methodology	Scales of analysis		Key Parameters	Cooling Mechanism	Cooling Effect	Assessment Methodology	Conclusions
Crown Structure	Spectral analysis of tree crowns	Individual	Crown Level	Infrared Radiation Properties, Canopy Structure	Radiation cooling	Canopy structure affects radiative properties, infrared radiation dominates cooling	Spectrometer measurements of canopy transmittance and reflectance	Infrared radiation dominates, canopy structure is a key influence
	Remote Sensing, Statistical Analysis	Group	Urban Forest Canopy	Urban Forest Canopy Cover, Patch Characteristics (Area, Perimeter, Shape)	Shade, Transpiration	Larger, tighter canopy patches cool better	Surface temperature extraction, correlation and regression analysis	Urban forest canopy cools, patch size and shape influence the effect
	Terrestrial Laser Scanning (TLS)	Individual	3D Canopy	Crown Size and Density, Leaf Optical Properties	Shade, Transpiration	Larger, denser canopies and specific tree species cool better	3D Point Cloud, Shade Analysis	Canopy structure and tree species affect cooling, larger canopies are more effective
Crown Density	Dummy-Variable Regression (math.)	Individual	Crown and species	Parameters of Plant Community Structure	Shade, Transpiration	Vegetation types with high canopy density and high porosity are more effective in cooling the air temperature	Field sample collection, Dummy variable regression analysis	Canopy density 0.81–0.85, porosity 0.31–0.35, plant community cooling effect is the best
	ENVI-met	Individual and Group	Single trees and groups of trees	Crown Density (LAI, LAD)	Shade, Transpiration	Higher density tree species are more effective in cooling	Field measurements, Analogue analysis	Dense canopy improves microclimate, best cooling at 3 pm
Crown Size	Thermal Imaging	Individual	Tree characteristics	Tree Characteristics (LAI, Crown Width, Leaf Color)	Shade	Wider canopies and higher LAI cool the surface more, with effects varying by surface type	Thermal Imaging Camera Measures Surface Temperature	Tree characteristics affect shade cooling, with the strongest cooling in the center and west of the shade shadow
	ENVI-met	Group	Residential trees	Crown Structure (Crown Diameter, Shape), Percentage Cover	Shade, Transpiration	Cooling increases with cover and is most effective for specific crown sizes	Air Temperature, Humidity Simulation	Canopy structure and cover influence air temperature, with a non-linear relationship

Table 3 (continued)

Research	Research Methodology	Scales of analysis		Key Parameters	Cooling Mechanism	Cooling Effect	Assessment Methodology	Conclusions
Crown Properties	Field Measurements, Questionnaires	Individual	Crown Properties	Crown Attributes (Crown Diameter/Branch Height Ratio, Leaf Clearance)	Shade	Higher crown diameter/branch height ratio and lower leaf clearance improve cooling and comfort	Mean Radiant Temperature (T _{mrt}), Thermal Sensory Vote (TSV)	Larger, denser canopies enhance thermal comfort, with crown diameter/branch height ratios being critical
Vertical Structure of the Tree Crown	Drones, Thermal Imaging, TLS	Group	Forest level	Canopy Structure (Vertical, Horizontal)	Shade, Transpiration	Evergreen broadleaf forests cool best, structure influences vertical temperature distribution	Multi-instrument measurement of temperature and structural parameters	Canopy structure affects the thermal environment, with broad-leaved evergreen forests having the best effect
	Air Temperature Measurement	Individual	Tree species	Tree species	Shade, Transpiration	High transpiration trees cool more and perform better in hot weather	Vertical Air Temperature Gradient Measurement	Tree Species and Weather Affect Cooling
Summary	Multiple methods integration	Individual to landscape scales	Crown size, density, structure, LAI, LAD, species	Primarily shade and transpiration	Shade, Transpiration, Radiation	Size, Density, Complexity, Diurnal Peak	Multi-scale assessment recommended	Canopy structural traits are critical determinants of cooling efficacy, with optimal ranges existing for maximum benefits

(2) Vertical structure of trees

The vertical structure of trees is also one of the influencing factors of tree cooling. A related study analyzed the cooling effect of seven urban tree species by measuring the temperature and humidity data of 100 trees and the parameters of tree height, crown height, and trunk circumference. That study found that vertical structural characteristics such as tree species, tree height, and crown height affected the cooling effect, of which the trunk circumference was a key indicator for evaluating the climate regulation service. This study recommended that core traits such as trunk girth be incorporated into green space planning and management to optimize tree species selection and to enhance ecological service benefits, providing scientific support for mitigating the UHI effect (Helletsgruber et al. 2020).

Some studies have also used ground-based LiDAR technology to quantitatively analyze the relationship between shading, vertical structure, planting environment, type of street trees, and thermal comfort improvement. They calculated the Universal Thermal Climate Index (UTCI) of each street tree, conducted multiple regression analysis, and found that the cooling effect of street trees with a larger crown volume is stronger, while the improvement of thermal comfort is weaker with an increase in diameter of the chest, and the thermal environment regulation is more advantageous for trees planted in a south direction. Trees planted in the south direction were found to be more advantageous for thermal environment regulation. The importance of the vertical structure of street trees on thermal comfort improvement was emphasized, and systematic management measures were called for, in order to optimize the planning

and management of urban street trees to enhance thermal comfort (이수빈, 최혜영, 조현길, 2021).

Some studies have even simulated the cooling effect of green space configuration through the ENVI-met model, verified it by comparing it with the measured data, and found that the vertical structure parameters such as leaf area index, crown width, and tree height have an impact on the cooling effect. The optimized model can provide accurate predictions for high-temperature microclimate mitigation and can guide urban planners to give priority to selecting tree configurations with high Leaf Area Index (LAI), wide crown width, and reasonable tree heights. These selections can effectively enhance the thermal environment regulation ability of green spaces (Li et al. 2023).

(3) Shade (of Tree)

As one of the most basic principles of tree cooling, the shading effect generated by tree canopies plays a crucial role in reducing surface and air temperatures. A small-scale experiment by Rahman et al. (2021) quantified the cooling effect of tree shade, reporting an average air temperature (AT) reduction of 2.1 °C during humid periods and 3.8 °C during dry periods. The study also found that shade lowered the Physiological Equivalent Temperature (PET) by up to 15 °C, demonstrating its significant role in alleviating the urban heat load, especially under arid conditions. It should be emphasized that shade acts as a cooling mechanism produced by tree canopies, rather than a condition affecting the trees themselves. The shading capacity depends on canopy density, leaf optical traits, and structure, which jointly determine the extent of temperature reduction.

Urban canopy shade systems can reduce daytime pavement temperatures and can improve thermal comfort by virtue of their durability properties, providing an optimized pathway for mitigating the UHI effect based on natural shading (Cheela et al. 2021). Studies using mean radiant temperature (T_{mrt}) as an indicator of heat stress have found that tree shading reduces daytime T_{mrt} in sparsely vegetated areas, and increased tree cover effectively mitigates high temperatures and heat stress risk (Lindberg et al. 2016). Scale outdoor experiments measuring evapotranspiration and shade in street canyons indicate that shading is the primary driver of cooling, with evapotranspiration playing a secondary role (Namazi et al. 2024; Wu et al. 2024).

The influence of shade can be evaluated vertically (canopy → sub-canopy → ground) and horizontally (extent of cooled area from trunk to surrounding surfaces) using field measurements and regression analysis. Shaded canopies reduce sub-canopy air and soil temperatures by lowering incoming shortwave radiation and increasing latent heat flux. Leaf brightness, density, thickness, and surface texture

influence cooling through albedo, radiation absorption, boundary-layer resistance, and transpiration capacity (Lin and Lin 2010). A schematic illustrating vertical and horizontal evaluation has been added in Fig. 4.

Canopy coverage is the main driver of daytime cooling, improving thermal comfort, whereas night cooling is weaker due to the heat island effect and requires complementary measures (Gillerot et al. 2024). The cooling effect of trees through shade is non-linear; vegetation density negatively correlates with temperature, and once coverage surpasses a threshold, the cooling effect is amplified. Thus, judiciously increasing tree cover is an effective approach to alleviate urban heat load (Rivera 2023).

However, this nonlinear amplification is not a universal rule, and several studies report contrasting findings. In some cities and at low-to-moderate coverages, a temperature reduction exhibits an approximately linear relationship with the increase in vegetation. For instance, some research found that for every 10% increase in vegetation coverage, the Land Surface Temperature (LST) consistently decreased by approximately 0.6–1.2 °C (Chen et al. 2025).

Crucially, other studies found that the cooling effect reaches a saturation point or threshold, beyond which the marginal benefit of increasing cover is diminished or plateaus. For example, a study in Wuhan found that the cooling effect reached a saturation point when the tree cover percentage reached approximately 50–60% (Ke et al. 2021), with very limited additional cooling observed beyond this level. Furthermore, for air temperature reduction, optimal benefits are typically observed within a Leaf Area Index (LAI) range of 3–5; beyond this range, the cooling efficiency may begin to decrease due to dense canopies restricting air circulation, potentially trapping longwave radiation, and reducing transpiration capacity per unit leaf area [73.74].

Therefore, future research needs to critically discuss the specific functional form of the relationship between tree cover, LAI, and actual cooling effects across different climatic zones, urban structures, and vegetation types to identify the optimal cover configuration.

3.6.2 Physiological property

Physiological traits directly govern evaporative cooling. A study by Gillner et al. (2015) demonstrated a strong correlation between transpiration rates and cooling magnitude. Species with high stomatal conductance ($> 150 \text{ mmol m}^{-2} \text{ s}^{-1}$) and leaf area density ($> 3.0 \text{ m}^2 \text{ m}^{-3}$) reduced air temperatures by 1.5–3.0 °C more than species with lower physiological activity. For instance, *Tilia cordata* achieved a 2.8 °C cooling effect, primarily driven by its high transpiration rate, whereas *Acer platanoides* with a lower rate provided only 1.2 °C of cooling under the same conditions. Urban

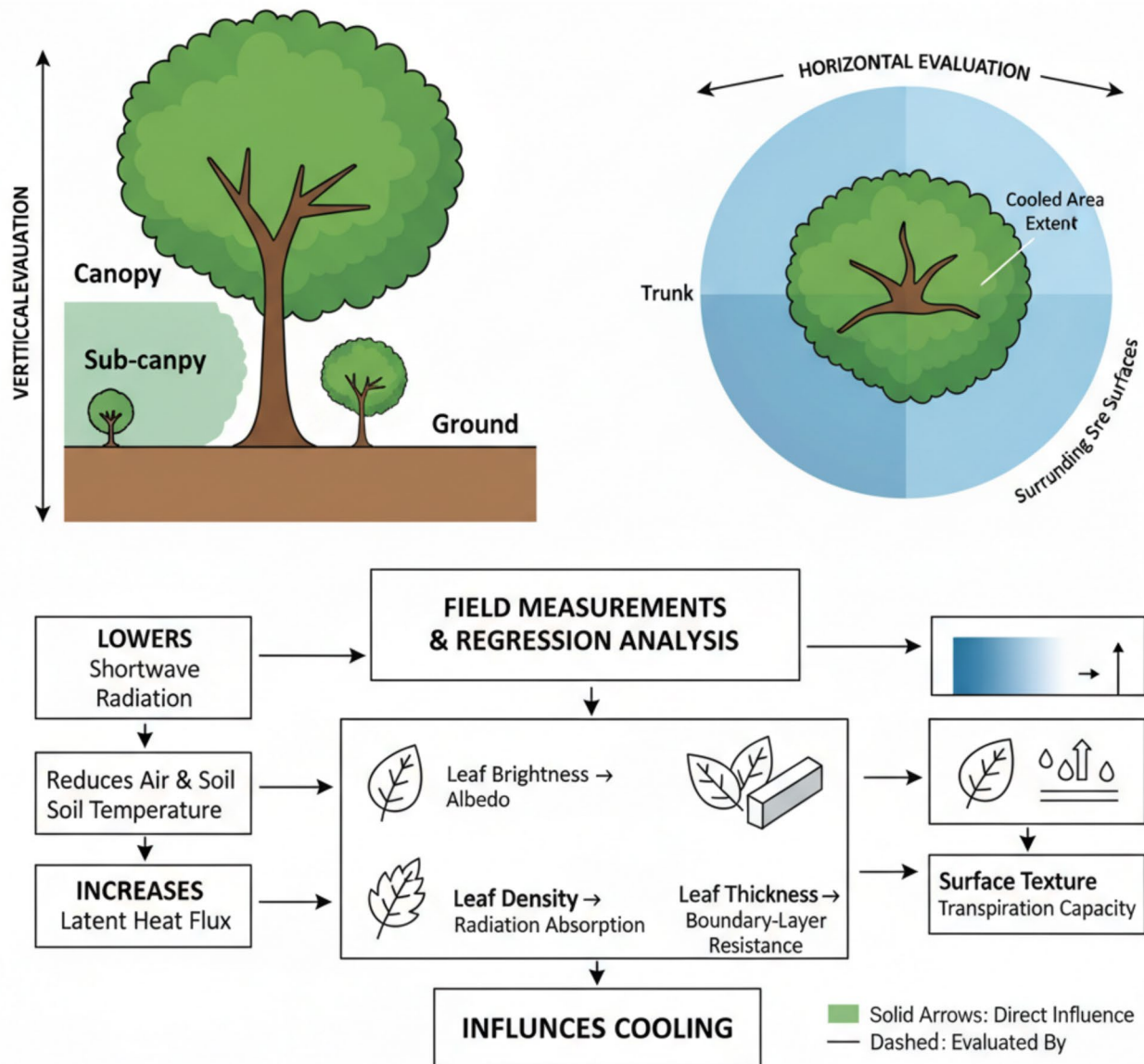


Fig. 4 Schematic Illustrating Vertical and Horizontal Shade Evaluation

greening should prioritize tree species with high cooling potential, in order to effectively mitigate the heat island effect (Gillner et al. 2015). The transpiration process can be incorporated into the CitySim modeling tool and linked with the CIM model to assess the impact of vegetation on microclimate regulation. Additionally, field verification can demonstrate that ground evaporative cooling effectively mitigates the intensity of the local heat island, thereby supporting high-temperature mitigation strategies that rely on vegetation transpiration (Mauree et al. 2019).

Tree transpiration facilitates heat removal, thereby lowering environmental temperatures. Consequently, several studies have employed heat dissipation probes to assess sap flux density in tree xylem, aiming to examine the effects

of tree transpiration on the environment, particularly in the context of varying water stress conditions. Sap flux densities and peak times of tree transpiration were elevated during the rainy season, as compared to the dry season. Transpiration was a dominant process year-round, contributing to notable air cooling through heat uptake. Transpiration alleviates water stress and regulates local climate, serving as a foundation for ecological management and climate-adaptive silviculture (Zhu and Ping 2013). Measurements of tree information across various street tree species indicate variability in transpiration water consumption. Additionally, a positive correlation exists between stomatal conductance and transpiration pressure difference, while gas exchange efficiency influences growth rate. The design of

tree trenches that limits soil moisture can hinder tree growth and diminish cooling effects. Optimizing the structure of tree trenches, such as by improving water retention, may enhance the potential for transpiration cooling. This serves as a critical consideration for the selection and design of urban street trees (Caplan et al. 2019).

3.6.3 Age and species

(1) Age of trees

The cooling capacity of trees is influenced by their age. Empirical research indicates differences in germination time, leaf discoloration time, and growing season length between trees in areas with low UHI intensity and those in areas with medium to high UHI intensity. The cooling performance of trees is affected by their characteristics, with tree age serving as the most reliable indicator of cooling capacity. Specifically, tall, mature trees exhibit considerable cooling capacity in areas with elevated heat island temperatures (Stanley et al. 2019).

(2) Tree types and species

The cooling efficiency of trees is contingent upon the specific type and species of the tree. This study evaluates the influence of various vegetation types on the thermal environment in mixed functional areas using a computational fluid dynamics (CFD) model. It demonstrates that tree characteristics, such as canopy morphology and density, play a role in regulating pedestrian thermal comfort during the daytime. Furthermore, the arrangement of these trees effectively lowers the average radiant temperature, offering scientific evidence for strategies aimed at mitigating the heat island effect and enhancing greening plans to address extreme high temperatures. Layout can decrease the average radiant temperature, offering empirical support for alleviating the heat island effect and addressing extreme heat conditions. (Sabrin et al. 2023).

The variation in plant area index among different tree species can result in distinct cooling effects provided by trees. The research on street trees has involved measuring the microclimate parameters of various tree species and estimating their physiological equivalent temperatures (PET). The findings indicated a positive correlation between the plant area index (PAI) and microclimate benefits, demonstrating that an increased PAI corresponds with enhanced pedestrian thermal comfort. Due to the substantial variation in PAI potential among tree species, it is advisable to identify high PAI species (e.g., those with dense canopies and appropriate leaf area distribution) in relation to the desired

canopy area in order both to enhance the shade cooling effectiveness of street trees and to improve urban thermal comfort (Sanusi et al. 2017).

Monitoring meteorological and transpiration data both within and outside the canopy reveals that tree species with a high leaf area index lower canopy temperature and shade physiological equivalent temperature through radiation interception and increased transpiration. Urban topography, including street orientation, building form, and green space layout, indirectly influences cooling by regulating wind flow. The selection of tree species with a high leaf area index, along with the optimization of ventilation corridors and green space connectivity, can improve the cooling efficiency of vegetation and alleviate extreme high temperatures (Rahman et al., 2020).

Various tree species exhibit differing rates of transpiration. This study assessed the transpiration and crop water stress index (CWSI) of 14 tree species. By analyzing the differences in canopy and air temperature, it was determined that conifers provide the most effective cooling effect under conditions of no water stress. In contrast, many broadleaf deciduous species such as *Quercus*, *Platanus* and *Populus* show reduced cooling capacity when experiencing water stress due to stomatal closure and decreased transpiration. The conclusion indicates that conifers are preferable in moisture-rich regions because of their efficient water utilization, while water-adapted tree species should be chosen in arid areas to preserve cooling effectiveness (Rana et al. 2020).

Related studies indicate that analyzing canopy morphology, plant area index (PAI), leaf morphology, and environmental parameters (such as temperature and humidity) reveals that tree species with broad canopies and moderate leaf density lower air and surface temperatures. Their cooling effect surpasses that of other types, highlighting their essential role in mitigating the UHI effect. Consequently, trees exhibiting these characteristics are recommended as urban cooling species (Ihsan and Rosline 2020).

Under extreme weather conditions, the variations in cooling effects among tree species become markedly pronounced. The research assessed canopy temperature and its correlation with stomatal conductance through thermal imaging, revealing notable variations in the cooling effects of various tree species on localized microclimates in urban settings during extreme heat. Park trees exhibited lower canopy temperatures as compared to street trees, conifers maintained temperatures near ambient levels, and small-leaved species proved to be cooler than their larger-leaved counterparts. The difference in leaf air temperature ($\Delta T(L-A)$) under elevated temperatures was influenced by stomatal conductance, which directly impacted cooling capacity. In extreme climates, therefore, it is advised to prioritize the selection and breeding of small-leaved species and to

optimize the arrangement of park and street trees to enhance thermal regulation benefits (Leuzinger et al. 2010).

A critical synthesis of tree traits reveals that cooling performance is a trade-off between morphological and physiological strategies. Morphological traits (e.g., high LAI, wide canopy) primarily govern shade-driven cooling, which is highly effective but can be limited by building shadows. Physiological traits (e.g., high transpiration rate) drive evaporative cooling, which is more sustainable in arid climates but constrained by water availability. No single species excels in all traits; therefore, species selection must be context-specific, balancing the desired cooling mechanism with local resource constraints (e.g., water) and urban form. Across the reviewed studies, trees typically reduced near-surface air temperature by 1–5 °C, surface temperature by 2–12 °C, and improved thermal comfort indices (PET, UTCI) by 5–15 °C equivalents, depending on canopy density, species, and local morphology.

4 Assessment methods of tree cooling

The UHI effect is becoming more pronounced due to both urbanization and the rising frequency of extreme heat events. Trees serve as a crucial component of urban green infrastructure, exerting a notable regulatory influence on

the thermal environment through mechanisms including shading (Yu et al. 2025). The research and evaluation methods employed in studies related to tree cooling exhibit greater diversity. Through a summary and analysis of pertinent research literature, four categories of research and assessment methods concerning tree cooling are identified: climate modeling and environmental simulation, remote sensing mapping, information extraction, and threshold and efficiency. These classifications elucidate the mechanisms by which trees regulate the thermal environment across multiple scales and dimensions, thereby offering a scientific foundation for tree cooling initiatives. Figure 5 provides an overview of the principal assessment approaches identified in the reviewed studies, including field measurements, remote sensing, and climate modelling.

4.1 Climate modelling and environmental simulation

The development of climate models and environmental simulations is a crucial research methodology for examining both the effects of trees on the climatic environment, including temperature, and the mechanisms through which trees facilitate cooling. A wide range of models has been applied, including City-LES 2.0, SLUCM-SEMM, WRF-BEP-CMAQ, and CFD, each with different strengths depending

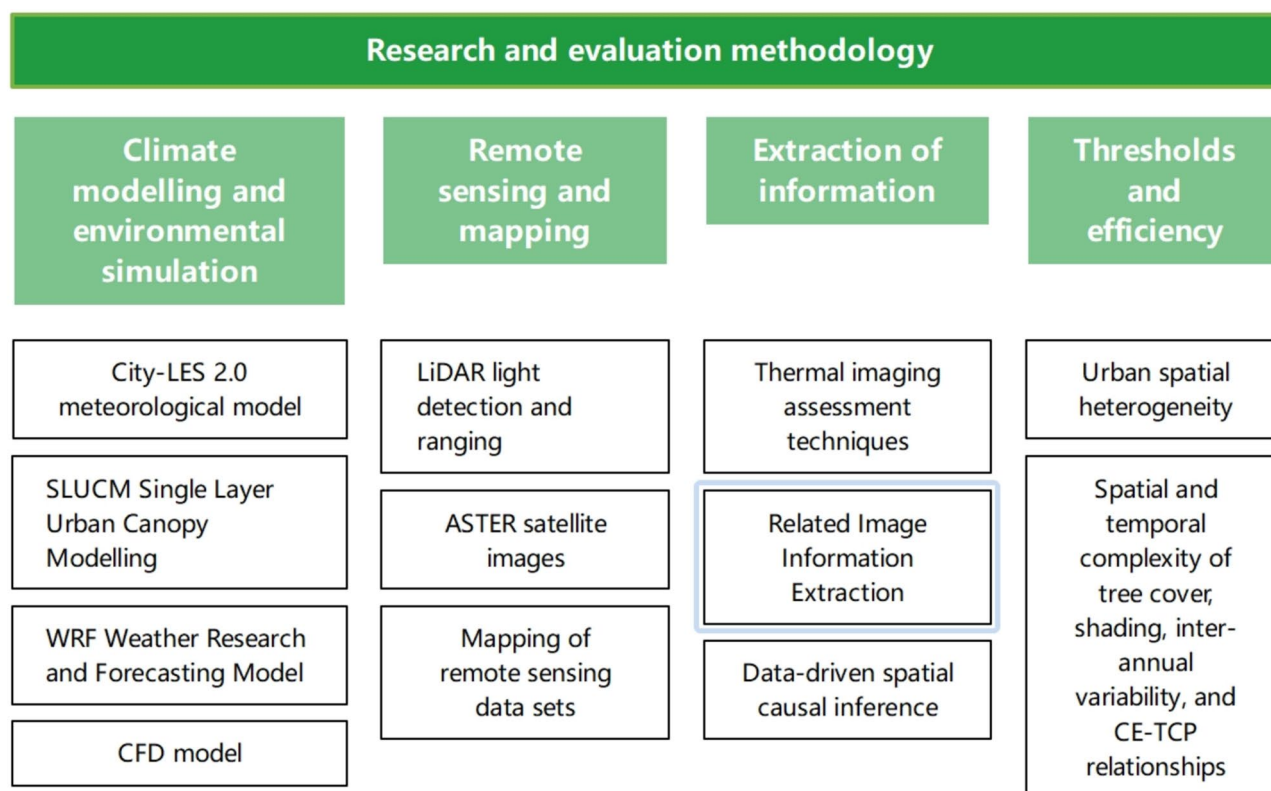


Fig. 5 Research and evaluation methods for tree cooling

on the spatial scale and research focus. The meteorological model (City-LES 2.0) simulates environmental climate at both city and urban neighborhood scales. It incorporates a method for light energy transfer in solar longwave and shortwave radiation, enabling a more realistic simulation of shading cooling methods, a quantification of the heat stress index, and a systematic evaluation of the cooling effects of trees (Kusaka et al. 2024). This model is particularly suitable for detailed microclimate analyses within neighborhoods due to its high-resolution LES framework.

The Single Layer Urban Canopy Model (SLUCM) was integrated with the Surface Energy Balance Model (SEMM) to quantitatively evaluate and forecast the influence of urban morphology, including trees and canopies, on the thermal performance of surfaces (Kumar et al. 2024). It is effective for intermediate-scale urban simulations where canopy and urban morphology interactions are critical. The Weather Research and Forecasting Model (WRF) was integrated with the Building Effects Parameterization (BEP) and the Community Multiscale Air Quality Model (CMAQ), both to develop an analytical framework capable of systematically analyzing and forecasting the effects of urban green infrastructure, including trees, on the city's meteorology, and to recommend specific management strategies (Paz et al. 2022). This model excels at city-scale climate forecasting and assessing broader urban heat mitigation strategies.

A CFD model utilizing a polyhedral grid of urban environments, incorporating geometrically explicit representations of trees, solar radiation, and the thermal impacts of tree transpiration, was implemented in a high-latitude city to validate and evaluate its efficacy in simulating and reproducing the microclimate (Brozovsky et al. 2021). CFD is best suited for detailed street-canyon or microclimate simulations, capturing fine-scale airflow and shading effects. The simulation of climate with and without trees, utilizing in situ measurements and computational fluid dynamics (CFD) within a street canyon, demonstrated that trees modified the standard diurnal cycle of surface and air temperatures in the canyon, influencing temperatures through mechanisms such as canopy transpiration (Pappaccogli et al. 2014).

Overall, while these methodologies are largely descriptive in the literature, a comparative evaluation highlighting which models are most appropriate for microclimate versus city-scale forecasting would enhance understanding of their relative strengths and applications. These methodologies for climate modelling and environmental simulation can be more intuitively illustrated to depict the varying impacts of trees on temperature across different factors. Their use can thoroughly regulate the impact of the natural environment, including airflow, sun radiation, and other variables affecting tree cooling research, to achieve more precise results. The remote sensing datasets and analytical techniques adopted across the reviewed literature are summarized in Table 4.

Table 4 Climate Modelling and Environmental Simulation for Tree Cooling Research

Model Name	Simulation Methodology	Simulation Scale	Scope of Application	Analogue Range(Clarified)	Specificities
City-LES v2.0	Atmospheric vortex simulation (LES)	Multi-scaling: urban- and neighborhood-scale simulations to discriminate against the micro-scale structure of buildings and trees	Urban Summer Heat Stress Mitigation Measures	Types of urban environments or climate conditions analogous to the target study area	Three-dimensional radiative transfer, output of multiple thermal stress indicators, GPU cluster operation
WRF+BEP+CMAQ	Domain Meteorological Models+Urban Parameterization+Air Quality Models	City/regional scale, year-round 1 km grid simulation	Such as Madrid in the Mediterranean climate	Climatic analogues: regions with similar climate and urban characteristics to the target city	Consideration of urban effects, multi-modal coupling, high resolution
CFD Model	3D RANS, Reynolds Stress Model Computational Fluid Dynamics	Fine urban geometry such as medium scale street canyons	High latitude cities in Northern Europe, cities with Mediterranean climate, etc	Urban geometry and climatic analogues: comparable street canyon layouts and meteorological conditions	Mesh Polyhedralization

4.2 Remote sensing and mapping

Remote sensing mapping is a prevalent method employed in the examination and evaluation of tree cooling. It utilizes images and data obtained from satellites and light detection, facilitating a broader scope of tree cooling research and enhancing quantification and modelling efficiency.

It involves employing Light Detection and Ranging (LiDAR) active remote sensing to acquire three-dimensional geometries of individual trees, which can be integrated with the PATH model to derive leaf area parameters (Leaf Area Index (LAI) and Leaf Area Density (LAD)) and to examine the influence of tree configuration on temperature regulation via urban climate modelling (Hu et al. 2018). LiDAR can capture the vertical structure of vegetation across extensive areas with high resolution, facilitating the prediction of understory temperature, and enabling the creation of high-precision thermal environment maps. Algorithms assess the influence of tree canopy on microclimate and illustrate the spatial distribution characteristics of temperature regulation by trees (Gril et al. 2023).

This work utilizes ASTER satellite imagery to extract data on urban surface temperature, buildings' footprints, and land use for planning; it aims to analyze fine thermal patterns and their spatial configuration to assess the impact of spatial structure on these thermal patterns. The generated research images can identify hotspots in urban temperatures, thereby facilitating the examination of the impact of tree-generated shadows on localized surface temperature patterns (Nath et al. 2021). The use of comparative spatial

modeling of land surface temperature (LST) and heat mitigation index (HMI) in urban landscapes, utilizing satellite imagery, can identify specific cold and hot zones within the city and can elucidate the contributing factors, including the influence of land use and tree vegetation on temperature dynamics (Jame et al. 2024).

A framework for cross-scale and reproducible analysis is developed, utilizing remote sensing technology to evaluate the efficacy of automatic mapping, classification, and thermal effect assessment of green infrastructure (GI). This framework integrates surface coverage parameters to systematically analyze the cooling effects associated with the type, quantity, and spatial arrangement of GI while quantifying its influence on land surface temperature (LST) through statistical modelling (Bartasaghi-Koc et al. 2019). The remote sensing dataset can be utilized to create a distinct tree dataset for precise analysis of spatial features, including tree position, height, canopy volume, and biomass, thereby offering high-resolution foundational data for evaluating the cooling efficacy of vegetation (Ma et al. 2023). The principal remote sensing methods, data sources, and parameters used for tree cooling analysis are summarized in Table 5, which provides a concise overview of how different sensing techniques support quantitative assessment and spatial modelling.

Furthermore, Table 6 presents a comparative summary of analytical methods for assessing tree cooling effects, highlighting their spatial scales, major advantages, limitations, and typical application scenarios to aid in selecting appropriate modelling tools.

Table 5 Summary of research methods for remote sensing mapping of tree cooling

Remote Sensing Methods	Satellite Data Sources	Spatial Resolution	Time Resolution	Core Parameters	Main Application Scenarios
Terrestrial Laser Scanning (TLS)	Un-	Un-	Un-	Leaf area index (LAI), Leaf area density, Path length distribution	Leaf area measurements of trees for microclimate regulation, Heat island effect, Building cooling, Air quality improvement, Ozone generation
Aerospace Lidar (LiDAR)	Un-	1–10米	Arbitrary (hours, days, seasons)	Vegetation Area Index (PAI), Vertical Complexity Index (VCI), Surface Cover Fraction, FRAGSTATS metrics	Green infrastructure classification, Climate studies, Thermal assessment, Ecosystem services, Greening interventions, Tree management
Thermal infrared remote sensing, Supervised classification	ASTER, QuickBird	ASTER: 90 m; QuickBird: 2.8 m	Aug-Sept 2002	Surface temperature, Urban components, Building attributes, Tree census data	Urban thermal modelling studies, Sustainable planning, UHI mitigation, Public health
Satellite imagery measurement LST, HMI spatial modelling	Not Specified	Not Specified	1993 and 2020	LST, HMI, Land Use, Building Density, Wet Vegetation, Watersheds	Locating hot/cold regions, spatial and temporal evolution of UHI, sustainable development and mitigation

Table 6 Comparative summary of analytical methods for assessing tree cooling effects

Method	Spatial Scale	Strengths	Limitations	Typical Application
ENVI-met	Micro-climate scale	Highly detailed simulation	Time-consuming; requires high-precision input	Neighborhood design evaluation
LiDAR	Tree crown structure scale	High accuracy and 3D capability	High cost; data processing complexity	Canopy volume estimation
Satellite imagery	Urban scale	Strong regional comparability	Limited spatial resolution	Urban heat island monitoring

4.3 Extraction of information

In the research and evaluation of tree cooling, the use of a number of different information extraction approaches is essential for examining distinct facets. Utilizing thermal imaging technology to quantify the transpiration cooling efficacy of vegetation, alongside the use of empirical measures to validate the feasibility of evaluating the cooling and dehumidification impacts of trees, helps to clarify the potential for addressing their thermal environment regulation advantages (Boldrin et al. 2019).

Unlike satellite-based remote sensing, information extraction methods rely on ground-level or human-scale data sources such as street-view images, pedestrian videos, and causal inference analysis. They focus on detailed, context-specific interactions between trees, built surfaces, and human thermal experience, whereas remote sensing emphasizes large-scale land surface temperature patterns. These techniques complement remote sensing by capturing micro-scale variations in tree cooling and human thermal perception that cannot be resolved by satellite imagery. Essential image data can be obtained from pedestrian video recordings of pertinent routes and integrated with street view imagery and microclimate information to develop a GVI 3D greening resolution model, then adjusted for satellite NDVI vertical resolution discrepancies, and precisely measured the spatial distribution of street trees in relation to local temperature interactions, thereby facilitating more accurate conclusions regarding the correlation between urban tree greening and ambient air temperature (Hong et al. 2019).

The effectiveness of urban greening, particularly trees, in mitigating heat stress can be evaluated through data-driven spatial causal inference utilizing hypothetical interventions. This method offers accessible data that can be broadly implemented to deliver targeted greening solutions for alleviating heat stress (Calhoun et al. 2024). The utilization of thermal imaging, pedestrian video, and data-driven spatial causal

inference facilitates the extraction of pertinent tree-temperature data and their respective correlations, hence enhancing both access to and application of research information for comprehensive investigations on tree cooling. Therefore, this section addresses fine-scale, human-centered analyses that cannot be captured by satellite data.

4.4 Thresholds and efficiency

The efficiency of tree cooling is usually affected by canopy cover, spatial diversity, and other factors. The accuracy of the calculation of tree cooling efficiency is also affected by factors such as land cover type, and it is lower than the actual value. The study comprehensively evaluates the cooling effect induced by trees, analyzes the influence of spatial differentiation effect on the efficiency, and innovatively constructs a revised tree cooling efficiency (RTCE) model which can more accurately reflect the tree cooling effect and can reduce the underestimation of tree cooling potential and cooling efficiency (Leng et al. 2024).

Tree cooling efficiency is also different in different urban densities. The study measured the relationship between green cover and the cooling effect of greening, simulated different combinations of densities and tree cover and verified them in the ENVI-met model, and analyzed the changing patterns of cooling efficiency in different time periods to derive the Tree Cover Ratio (TCR) with the highest cooling efficiency; this provides a science-based recommendation for the green cover threshold (Ouyang et al. 2020).

Tree coverage and tree shading can improve the cooling efficiency of trees. The study selected several typical tree patches to study microclimate heterogeneity and urban surface type differences in the cooling effect of tree shading, and it found that tree shading can improve the local thermal environment and that the external thermal stress affects the cooling efficiency of trees. This enriches the understanding of the cooling mechanism of tree shading and reveals the moderating effect of the environmental conditions on the efficiency of shading (Jiao et al., 2024).

Regarding further studies on the relationship between tree cover (TCP) and cooling efficiency (CE), the spatial and temporal complexity of the CE-TCP relationship can be investigated using satellite-derived MODIS observations. A nonlinear CE-TCP relationship with an initial sharp decrease in CE followed by a gradual decrease with increasing TCP exists globally and is more pronounced in tropical and arid climates (Zhan et al. 2024).

And the efficiency of tree cooling is also affected by the inter-annual variation of tree cover. This study monitors the long-term analysis of global urban tree cooling efficiency (TCE) based on multi-satellite data and finds that there are interannual differences in TCE, with low fluctuations in high-coverage cities and

limited enhancement of the cooling capacity by vegetation expansion, that reveal the association between the level of tree cover and the variation of TCE and provide scientific evidence for the greening strategy of the city (Leng et al., 2024).

Tree cooling also has an obvious threshold; the higher the tree cover, the higher the tree cooling rate. The study captured information on the increase and decrease of green space, including trees, by constructing the UG dynamic index (UGDI) combined with diurnal temperature and humidity data to analyze the association between green space dynamics and temperature change, revealed the mechanism of the influence of trees on the threshold of cooling, supported the dynamic regulation of the thermal environment, and revealed the important influence of the dynamic process of green space change, including trees, on the threshold and efficiency of cooling. Overall, studies consistently indicate that tree cooling efficiency thresholds are primarily determined by canopy density, tree height, and surface albedo, with typical temperature reductions ranging between 1 °C and 4 °C under urban summer conditions. Across these multidimensional urban settings, a general pattern emerges: tree cooling is most effective in moderately open environments with balanced canopy cover (30–50%), while efficiency declines in both highly dense and sparsely vegetated areas, underscoring the need to align planting strategies with urban form and climatic context (Wang and Zhou 2022).

5 Cooling Measures and Design

This chapter systematically constructs an integrated system of tree cooling technology, forming a framework for the implementation of the whole chain of ‘planning-tools-strategies-scenarios’, optimizing the shading layout of street trees through microclimate simulation and spatial analysis technology, and strengthening the effectiveness of local thermal regulation. The integration of GREENPASS(R), ENVI-met, and the intelligent analysis platform builds a quantitative assessment toolset for the thermal environment and supports the analysis of thermal conflicts in multiple scenarios. In addition, the innovation of the TUNEE model drives the selection of tree species and mixed planting strategies, which is combined with passive irrigation to achieve the efficiency of temperature reduction in phases. Based on the multi-scale study of vertical cooling gradient, canopy configuration, and spatial heterogeneity, studies revealed the differentiated regulation mechanism of residential and park scenarios, and we will promote the synergistic optimization of the precise management of thermal environment and ecological benefits. Figure 6 illustrates the multiscale design framework for tree-based cooling measures, covering planning principles, planting strategies, and tool-based simulations.

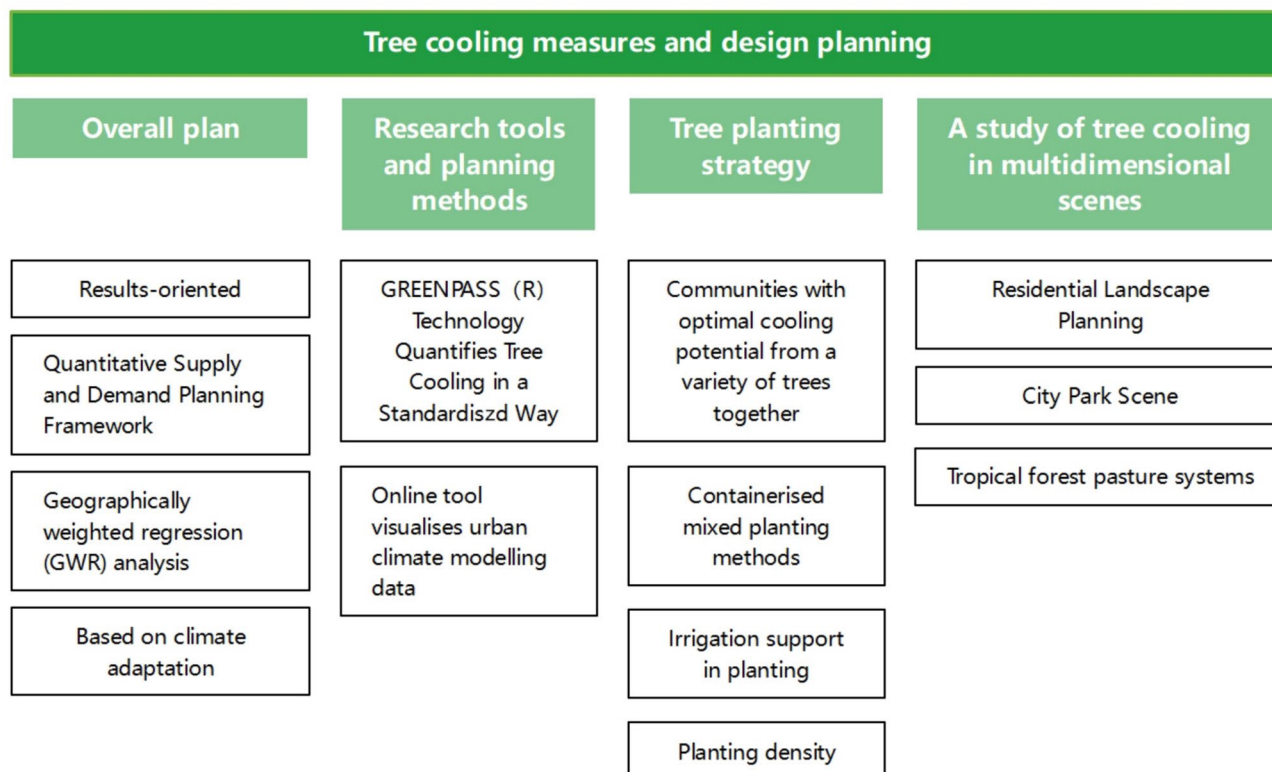


Fig. 6 Tree Cooling Measures and Design Planning

5.1 Overall plan

Effective utilization of the cooling impact of trees necessitates deliberate urban scientific masterplanning. A study utilizing microclimate-GIS technologies to simulate and optimize climate-adaptive landscapes demonstrates that street tree shading accounts for 67% of the cooling effect. A climate-focused planning framework was developed to incorporate the ongoing arrangement of canopy and tree species adaptation techniques, creating an ecological design approach to alleviating the heat island effect (Teoh et al. 2022).

From the standpoint of supply and demand, certain studies amalgamate remote sensing, meteorological, and socioeconomic data to develop a quantitative framework for the supply and demand of urban forest temperature-control services (TRS), categorizing four types of spatial matching modes to precisely identify hotspot areas that exhibit an imbalance between supply and demand. Differentiated forest allocation strategies are suggested, based on zoning characteristics, to enhance the efficacy of geographical distribution of cooling services and to offer a scientific foundation for planning and design (Li et al. 2022).

A separate study uses geographically weighted regression (GWR) to examine the impact of green space quantity and fragmentation on land surface temperature (LST) and uncovers the spatial variability in the cooling efficacy of trees. Differentiated tree arrangement techniques are recommended to develop management schemes for fragmented green spaces in urban and rural areas, and a robust thermal environment management framework is established to facilitate precise cooling decisions (Wang et al. 2024).

Additionally, much research utilizes the Sustainable Development Goals (SDGs) as a framework for urban design, incorporating ENVI-met-PET simulation technology and developing a climate-adaptive urban design verification system. Optimizing green cover can enhance transpiration cooling benefits by 23%, positioning the strategic placement of trees as essential for regulating thermal environments and offering a quantitative framework for sustainable urban cooling (Ahmed et al. 2024).

5.2 Research tools and planning methods

The advancement of research instruments and the recognition of novel planning scenarios is a transformative catalyst for enhancing design and planning to optimize tree cooling efficacy. The GREENPASS(R) technique quantifies the influence of trees on urban climate in a standardized manner, facilitating comparative studies with the ENVI-met(R) model to optimize the cooling effect of trees (Kraus and Scharf 2019).

A separate study created an online climate visualization platform that incorporates decomposition methods to analyze biophysical effects and temperature response mechanisms for precise quantification of tree thermal regulation contributions. By employing multi-scenario thermal conflict simulation and scenario optimization, it transcends the constraints of the ENVI-met system to evaluate the impact of vegetation, thereby augmenting the decision support capacity of urban climate simulation (Wollschläger et al. 2022).

Despite significant progress in developing analytical tools and planning frameworks, a substantial research–practice gap remains in translating these scientific insights into operational urban forestry policies. Many planning tools, while technically advanced, are not effectively embedded into local governance mechanisms due to institutional fragmentation, inconsistent data standards, and the lack of cross-sector coordination. Bridging this gap requires collaboration between researchers, planners, and policymakers to co-develop standardized guidelines and operational frameworks that connect modeling results with on-the-ground management practices, ensuring that scientific evidence can inform real-world decision-making.

5.3 Tree planting strategy

The research on tree planting tactics encompassed the cooling potential of diverse tree communities, containerized mixed planting techniques, planting density, and irrigation support for planting. Research on tree planting strategies for urban park design aimed at maximizing cooling effects involved the collection of physiological properties of predominant tree species, the assessment of their physiological characteristics and their cooling efficacy through the Thermal Urban Environment Energy (TUNEE) equilibrium model, and the analysis of individual plant and density effects. Efficient tree species were evaluated, resulting in the ideal distribution of six tree species and 19 plants within a 50 × 50 m area, in order to establish a model for the thermal regulation of park vegetation (Barradas et al. 2022).

A study examined the differential cooling effects of various tree species and planting methods by conducting container cultivation tests while monitoring soil temperature, humidity, and physiological markers. The findings indicated that the mixed planting approach is superior and recommended the temporal sequence configuration strategy of "rapid initial shading by fast-growing species + gradual regulation by medium-speed species" to attain synergistic optimization of short-term cooling and long-term thermal regulation (Rahman et al. 2023).

The cooling effect of tree canopy varies in urban planning. The study demonstrates, via in situ climate measurements and morphological analyses, a negative correlation between the tree cover ratio (TCR) of pocket parks and the

UHI effect. It indicates that the cooling effect is maximized when the TCR is 42% or higher. Furthermore, it advocates for high-density urban areas to adopt multilayered vegetation structures and vertical greening systems, as well as to develop a threshold-guided precise greening configuration model to establish a quantitative design benchmark for mitigating the heat island effect (Lin et al. 2017).

In the context of tree planting, the research measured the growth and transpiration of trees and developed three types of planting pits with varying capacities for rainwater collection and internal water retention for comparative analysis. The findings confirmed that the use of a passive irrigation system, which channels rainwater to the trees, may enhance growth and transpiration while facilitating cooling through the transpiration process (Thom et al. 2022).

5.4 A study of tree cooling in multidimensional scenes

The function of trees in temperature regulation differs across various contexts. In residential settings, the study employed the ENVI-met model to examine the vertical cooling gradient of vegetation, comparing the thermal regulation effectiveness of trees, shrubs, and herbs. It concluded that tree canopy shade accounts for 58% of the enhancement of the local thermal environment; its three-dimensional cooling efficacy was found to be markedly superior to that of low-growing vegetation, thereby offering guidelines for landscape design in high-density developments based on vertical thermal dynamics. This offers directives for landscape design in high-density areas, informed by vertical thermal stratification (Liao et al. 2021).

In the examination of tree cooling within urban park contexts, structural equation modelling (SEM) was employed to elucidate the regulatory principles governing tree coverage and spatial configuration on the park cold island effect (PCII). The findings indicated that augmenting coverage and optimizing patch morphology (minimizing irregularity and dispersion) could collaboratively enhance the PCII, thereby offering a dual optimization strategy between the coverage threshold and spatial configuration for the design of the park cooling system (Liao et al. 2023).

The study gathered data on forest and pasture systems in tropical forest pastures, including tree cover, woody carbon stock, and local air temperature. It revealed a positive linear correlation between tree carbon stock and the cooling effect, with the magnitude of this effect being relatively constant. Additionally, it was found that increasing system coverage can simultaneously improve carbon sequestration and local cooling capacity, offering a natural solution for addressing climate change through the synergistic effects of sequestration and cooling (Zeppetello et al. 2022).

6 Implications and impacts

6.1 Environmental impact

Tree cooling can influence both environmental and social factors. A study utilizing the microscopic computational fluid dynamics (CFD) model ENVI-met has been done, to simulate thermal settings and analyze street trees and the influence of cooling on the urban ecosystem. The study demonstrated that street trees mitigate the thermal environment, highlighting their essential function in boosting thermal comfort and walkability (Jia and Wang 2021).

Urban trees can also significantly reduce outdoor heat, optimize thermal comfort and offset urban heat islands, improving environmental quality and enhancing climate resilience (Li et al. 2024b). The mitigation of the Urban Heat Island (UHI) effect by trees is well-documented. Synthesizing results from multiple studies (Jia and Wang 2021), tree canopies have been shown to reduce the Universal Thermal Climate Index (UTCI) by 1.5 to 4.0 °C on hot summer days. For example, Zhu and Kensek (2024) (Zhu and Kensek 2024) recorded a maximum reduction of 2.2 °C in downtown Los Angeles, while studies in European cities reported average UTCI reductions of 2.5 °C Yin et al. (2024a). Tree canopies contribute to the natural environment and climate by alleviating the urban heat island effect through shading and evapotranspiration, thereby regulating urban thermal environments and enhancing humidity; this collectively mitigates heat and supports ecological balance (Yin et al. 2024b).

Trees contribute to lowering indoor temperatures and to maintaining relative humidity, thereby enhancing the thermal comfort of buildings. A study employed a combination of on-site measurements and computer modeling to assess the impact of tree cover on indoor thermal comfort in residences by evaluating the urban heat island (UHI) effect and tree cover (Taleghani et al. 2019). The findings indicated that increased tree cover correlates with reduced temperatures, thereby improving indoor thermal comfort and offering solutions (Rachman 2015).

6.2 Social impact

Urban tree cooling influences social equity and economic efficiency. Studies reveal a consistent correlation between tree canopy distribution and socioeconomic status, showing that low-income and minority neighborhoods typically have less canopy cover and consequently experience higher heat exposure, intensifying environmental injustices. Tree cooling influences social equality and economic efficiency. This analysis examines social thermal sensitivity and tree cooling capability to highlight disparities in thermal

benefits resulting from the unequal distribution of tree canopy within the context of environmental justice. Vulnerable populations had low tree cover; augmenting this cover could synergistically improve ecological and socioeconomic benefits while fostering equitable greening techniques grounded in natural solutions (Zhou et al. 2021).

The cooling effects of trees vary across different areas, with one study indicating that inconsistent urban tree cover intensifies disparities in heat risk. The distribution of urban tree canopy (UTC) is frequently correlated with socioeconomic status, where regions with diminished UTC are generally found in older, lower-income neighborhoods. Consequently, these areas experience reduced access to cooling benefits, thereby intensifying environmental injustices (Zhuang et al. 2023).

Targeted canopy expansion in disadvantaged neighborhoods thus serves as both an equity intervention and health strategy, addressing critical cooling needs while fostering social and ecological co-benefits. Enhancing environmental justice involves improving urban ecosystems and promoting social equity. Tree planting in socially disadvantaged neighborhoods can provide cooling benefits while simultaneously addressing environmental justice and tree equity. Enhancing the tree canopy in these regions addresses critical cooling requirements and fosters social and ecological co-benefits, while also rectifying the distributional injustices linked to urban heat (Zhou et al. 2021).

However, large-scale greening requires careful implementation to avoid "green gentrification," where improved environmental quality may increase property values and displace existing residents, highlighting the need for approaches that integrate environmental and social justice. Affluent, predominantly white neighborhoods exhibit greater tree density and biodiversity, while marginalized areas experience a deficiency in green infrastructure, resulting in heightened health risks for minority communities, due to reduced tree cover. Tree cooling can address disparities in canopy cover among various ethnic and socioeconomic groups (Gregory and Ajibade 2024). Targeted greening initiatives can enhance cooling and health equity, promote environmental justice, and yield synergistic ecosocial benefits (McDonald et al. 2024). Furthermore, equitable access to tree cooling can reduce risks and can enhance a community's well-being.

Observations from space station thermal radiometer experiments have revealed negative correlations between household income and surface temperatures, indicating that low-income neighborhoods endure elevated temperatures due to inadequate vegetation cover and low evapotranspiration, with temperature disparities intensified by rising regional average temperatures (Yin et al. 2023). Tree

cooling can promote environmental justice by reducing thermal disparities in communities. Equitable distribution of tree canopies ensures that all communities, especially marginalized ones, receive these cooling effects, enhancing health and well-being while promoting biodiversity conservation (Locke et al. 2023).

While the cooling effects of urban trees bring substantial social benefits, it is also important to recognize potential unintended consequences. In some cities, large-scale greening projects have contributed to "green gentrification," where the enhancement of environmental quality and thermal comfort has simultaneously increased property values and displaced lower-income residents. This phenomenon highlights the need for equitable planning approaches that integrate environmental sustainability with social justice to ensure that the benefits of urban greening are accessible to all communities.

7 Conclusion

The main contribution of this paper is to propose an integrated multi-scale and multi-dimensional comprehensive analysis framework for tree cooling, synthesizing the understanding of the coupling mechanism between tree morphology, physiological characteristics and social and environmental benefits, and providing systematic references for urban thermal environment governance. This review comprehensively examines the multifaceted contributing elements, evaluation techniques, design strategies, and environmental and social advantages of urban tree cooling effects. Comprehensively examines the multifaceted contributing elements, evaluation techniques, design strategies, and environmental and social advantages of urban tree cooling effects. This work synthesizes research advancements regarding the cooling effects of trees from a systematic and interdisciplinary viewpoint and offers an in-depth analysis of the theoretical discourse on influencing factors, research methodologies, and assessment techniques related to tree cooling. This is crucial essential for formulating effective strategies to amplify the cooling effects of trees, and it provides a scientific foundation for the practical implementation of tree cooling initiatives and design planning, thereby supporting the development of sustainable urban development and ecological civilization.

Despite synthesizing a broad range of literature on urban tree cooling, this review has several acknowledged limitations. First, many reported cooling values are derived from short-term field experiments or simulation models, which may not fully capture the complexity of long-term seasonal and inter-annual variability. Second, inherent differences in

experimental design (e.g., sensor height and placement), climate classification, and land-cover types introduce substantial uncertainties during cross-study quantitative comparison. Third, the majority of available quantitative results were obtained from cities in temperate and subtropical climates, leaving the evidence base for arid, cold-climate, or high-altitude urban areas significantly limited. Finally, this review primarily focuses on biophysical cooling mechanisms, but the critical socioeconomic constraints on urban forestry—such as maintenance cost, water demand, shading conflicts, and public health concerns related to allergens—were not fully covered.

Future investigations into tree cooling should prioritize quantifying the enduring effects of tree cooling on urban microclimates and examining the viability of tree-based solutions in relation to global warming, as well as formulating context-specific tree cooling planning guidelines for planners, in order to maximize the potential of trees in alleviating urban heat (Gregory and Ajibade 2024). Moving forward, future studies should prioritize long-term monitoring and cross-climate comparison to fill geographical gaps. Additionally, integrating hydrological–physiological models will improve the reliability of cooling predictions. Ultimately, combining robust microclimate simulation with comprehensive socio-ecological analysis is essential to support more accurate and equitable urban planning decisions. Future research should focus on developing comprehensive models that can integrate remote sensing monitoring, urban microclimate simulation and socioeconomic data to evaluate the long-term cooling effects of different urban tree species and spatial patterns.

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Declarations

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